

# Lecture 03 — Image Filtering

## Lecture 3

Kaveh Fathian

Computer Science Department, Colorado School of Mines

# Image Filtering



# Learning Objectives

By the end of this lecture, you should be able to:

- Explain filtering as a local neighborhood operation.
- Distinguish correlation and convolution operations.
- Predict the behavior of common kernels.
- Explain linearity, shift invariance, boundary handling, and separability.
- Apply Gaussian, Sobel, sharpening, and median filters.
- Use correlation for basic template matching and recognize its limitations.

# Fundamental Equations of Computer Vision

## 1. Image Filtering

$$y[m, n] = \sum_{k, l} h[k, l] I[m + k, n + l]$$

## 2. Optical Flow

$$I(x, y, t) = I(x + \Delta x, y + \Delta y, t + \Delta t)$$

## 3. Camera Geometry

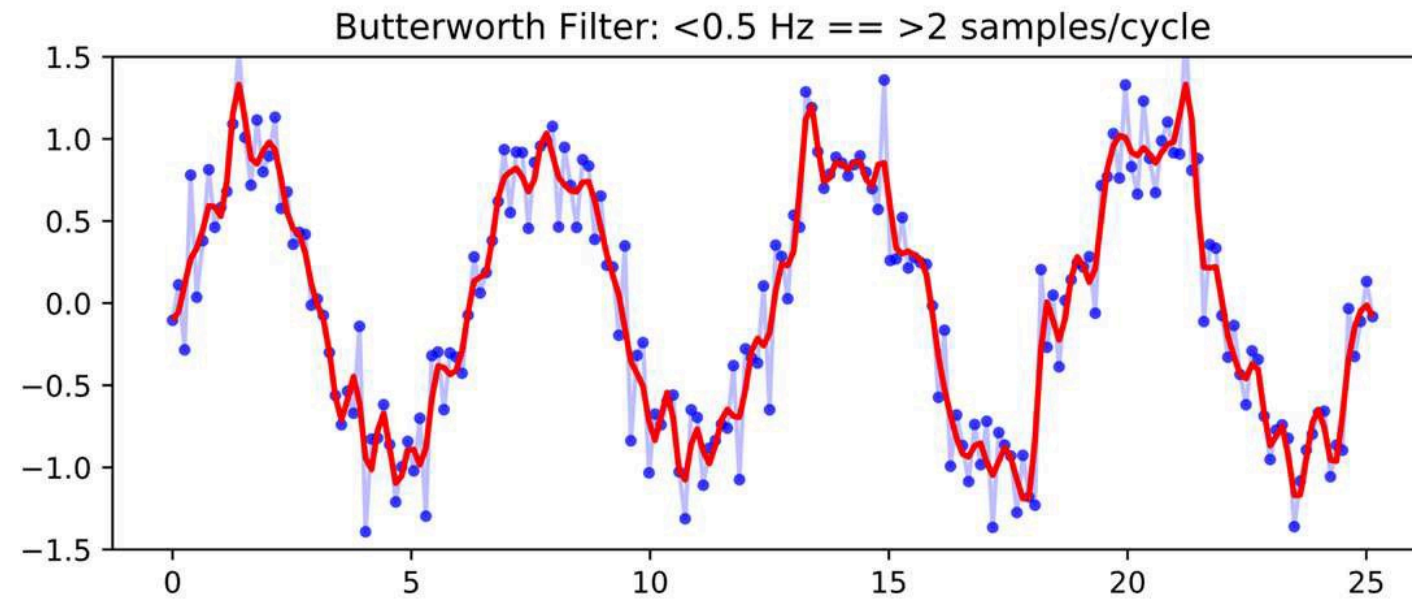
$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \mathbf{X} \quad \mathbf{x}^T \mathbf{F} \mathbf{x}' = 0$$

## 4. Machine Learning

$$\arg \min_{\mathcal{S}} \sum_{i=1}^k \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2 \quad y = \varphi(\mathbf{w}^T \mathbf{x} + b)$$



# Filtering Modifies a Signal



Noisy samples and a smoothed estimate

**Filtering** transforms a signal to:

- suppress undesired components such as noise
- emphasize useful structure
- extract specific features

Filters can operate on continuous or discrete signals in one or more dimensions.

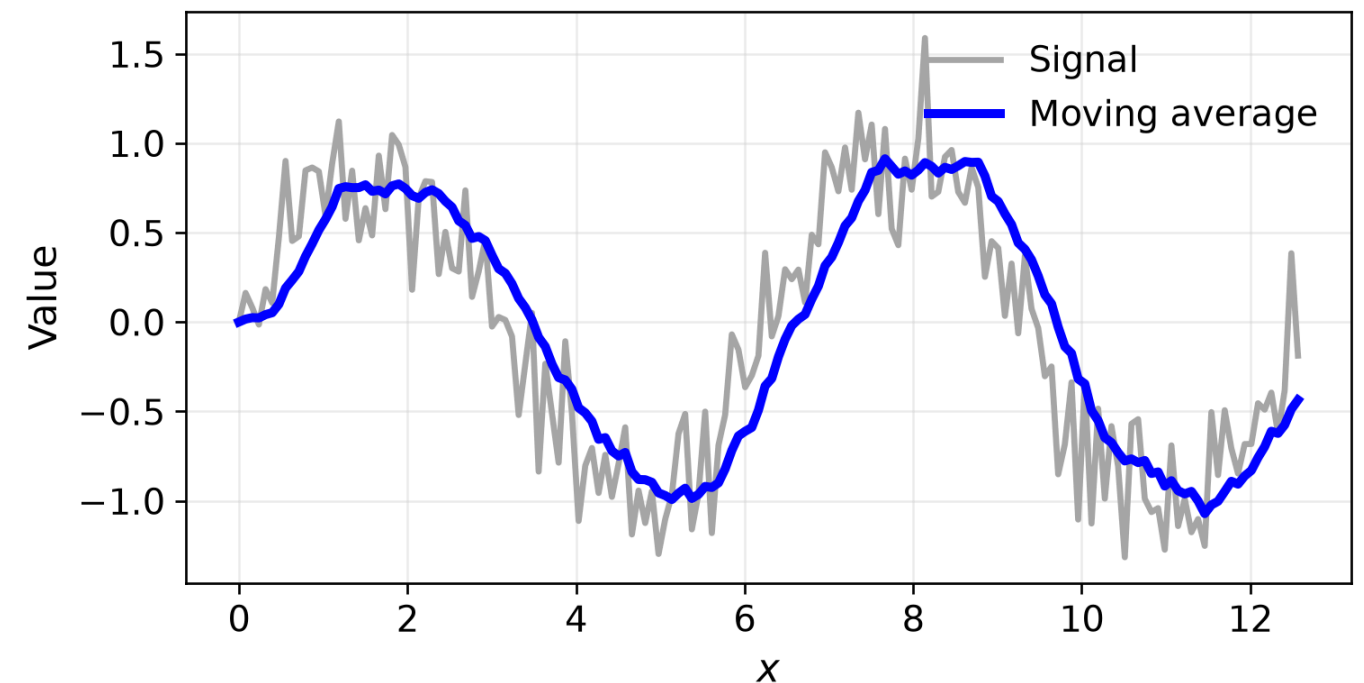
# 1D Moving Average Filter

For a window size  $k$ :

$$y[n] = \frac{1}{k} \sum_{i=n-k+1}^n (1)I[i] \quad \text{or}$$

$$y[n] = \frac{1}{k} \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} I[n-k+1] \\ I[n-k+2] \\ \vdots \\ I[n] \end{bmatrix}$$

```
1 x = np.linspace(0, 4*np.pi, 160)
2 rng = np.random.default_rng(7)
3 noisy = np.sin(x) + 0.28*rng.normal(size=x.size)
4
5 k = 10
6 ker = np.ones(k)/k
7 smoothed = np.convolve(noisy, ker, mode="full")
8
9 fig, ax = plt.subplots(figsize=(6.5, 3.5))
10 ax.plot(x, noisy, c="0.65", lw=2, label="Signal")
11 ax.plot(x, smoothed, c="blue", lw=3, label="Moving average")
12 ax.set(xlabel="$x$", ylabel="Value")
13 ax.legend(frameon=False); ax.grid(alpha=0.2)
14 plt.tight_layout(); plt.show()
```

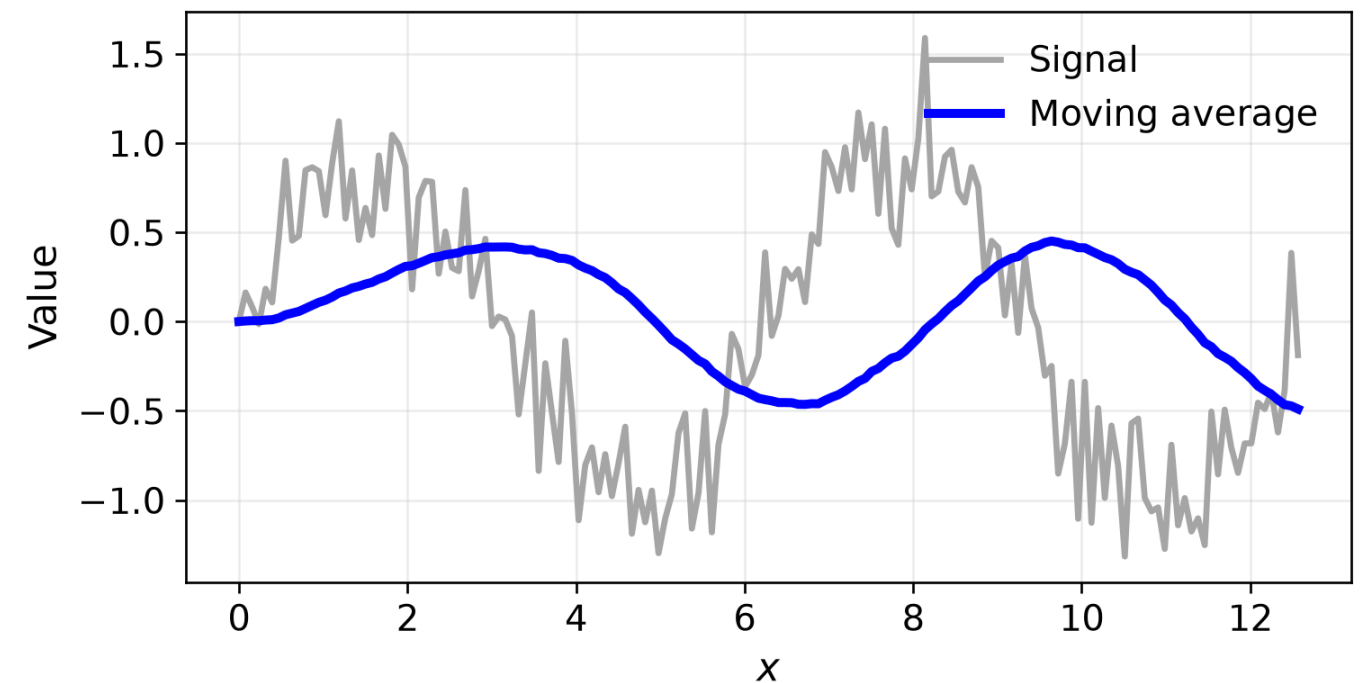


# The Window Size $k$

A larger averaging window:

- removes more rapid variation
- produces a smoother output
- spreads changes over a wider region
- can erase narrow but meaningful features

```
1 x = np.linspace(0, 4*np.pi, 160)
2 rng = np.random.default_rng(7)
3 noisy = np.sin(x) + 0.28*rng.normal(size=x.size)
4
5 k = 50
6 ker = np.ones(k)/k
7 smoothed = np.convolve(noisy, ker, mode="full")
8
9 fig, ax = plt.subplots(figsize=(6.5, 3.5))
10 ax.plot(x, noisy, c="0.65", lw=2, label="Signal")
11 ax.plot(x, smoothed, c="blue", lw=3, label="Moving average")
12 ax.set(xlabel="$x$", ylabel="Value")
13 ax.legend(frameon=False); ax.grid(alpha=0.2)
```

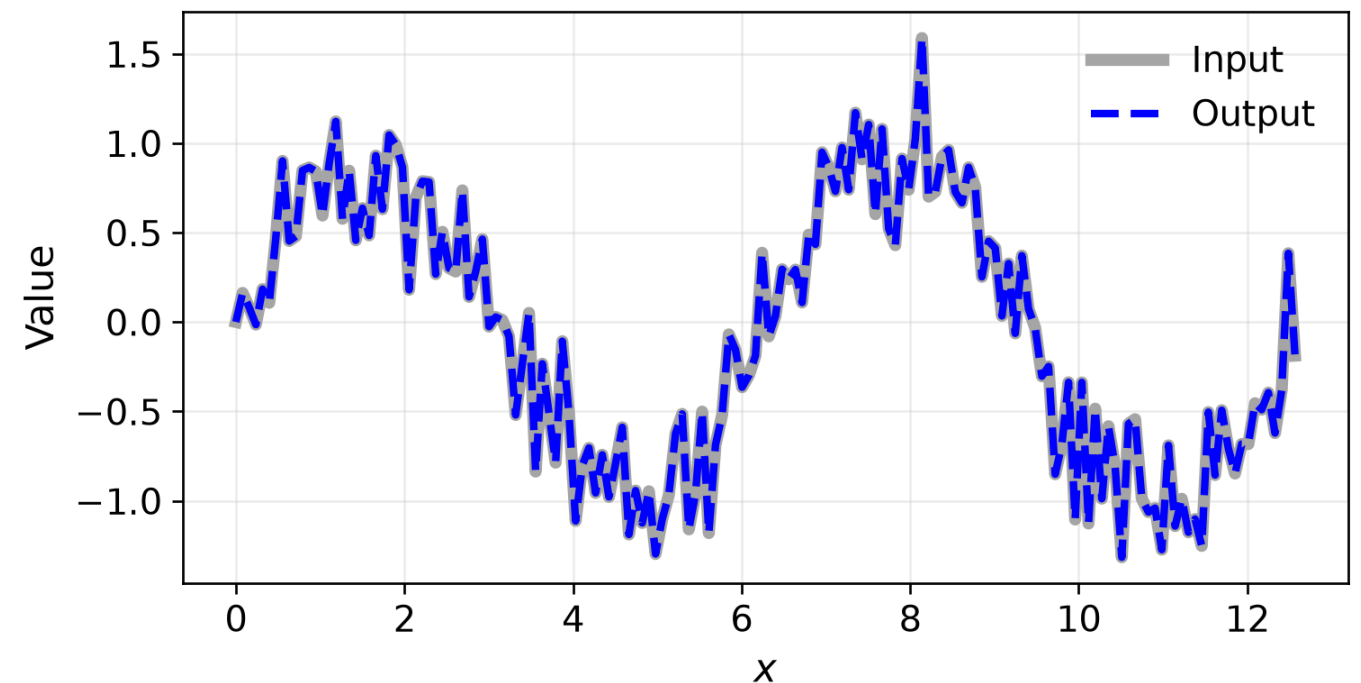


# Filter Kernel

What is this filter doing?

$$y[n] = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \end{bmatrix} \begin{bmatrix} I[n - k + 1] \\ I[n - k + 2] \\ \vdots \\ I[n] \end{bmatrix}$$

```
1 x = np.linspace(0, 4*np.pi, 160)
2 rng = np.random.default_rng(7)
3 noisy = np.sin(x) + 0.28*rng.normal(size=x.size)
4
5 k = 20
6 ker = np.zeros(k); ker[-1] = 1
7 output = np.correlate(np.pad(noisy, (k-1, 0)), ker)
8
9 fig, ax = plt.subplots(figsize=(6.5, 3.5))
10 ax.plot(x, noisy, c="0.65", lw=4, label="Input")
11 ax.plot(x, output, "b--", lw=2.5, label="Output")
12 ax.set(xlabel="$x$", ylabel="Value")
```



# Image Filtering

Image filtering computes a function of the local neighborhood at each pixel position:

$$y[m, n] = \sum_{k, l} h[k, l] I[m + k, n + l]$$

Filter window/kernel size: “ $k$ ” and “ $l$ ”

# Image Filtering

Image filtering computes a function of the local neighborhood at each pixel position:

$$\underbrace{y[m, n]}_{\text{output}} = \sum_{k, l} \underbrace{h[k, l]}_{\text{filter kernel weights}} \underbrace{I[m + k, n + l]}_{\text{neighboring image value}}$$

- $y[m, n]$ : output value at image coordinate  $(m, n)$
- $h[k, l]$ : filter coefficient at local offset  $(k, l)$
- $I[m + k, n + l]$ : input image value at the corresponding neighboring location

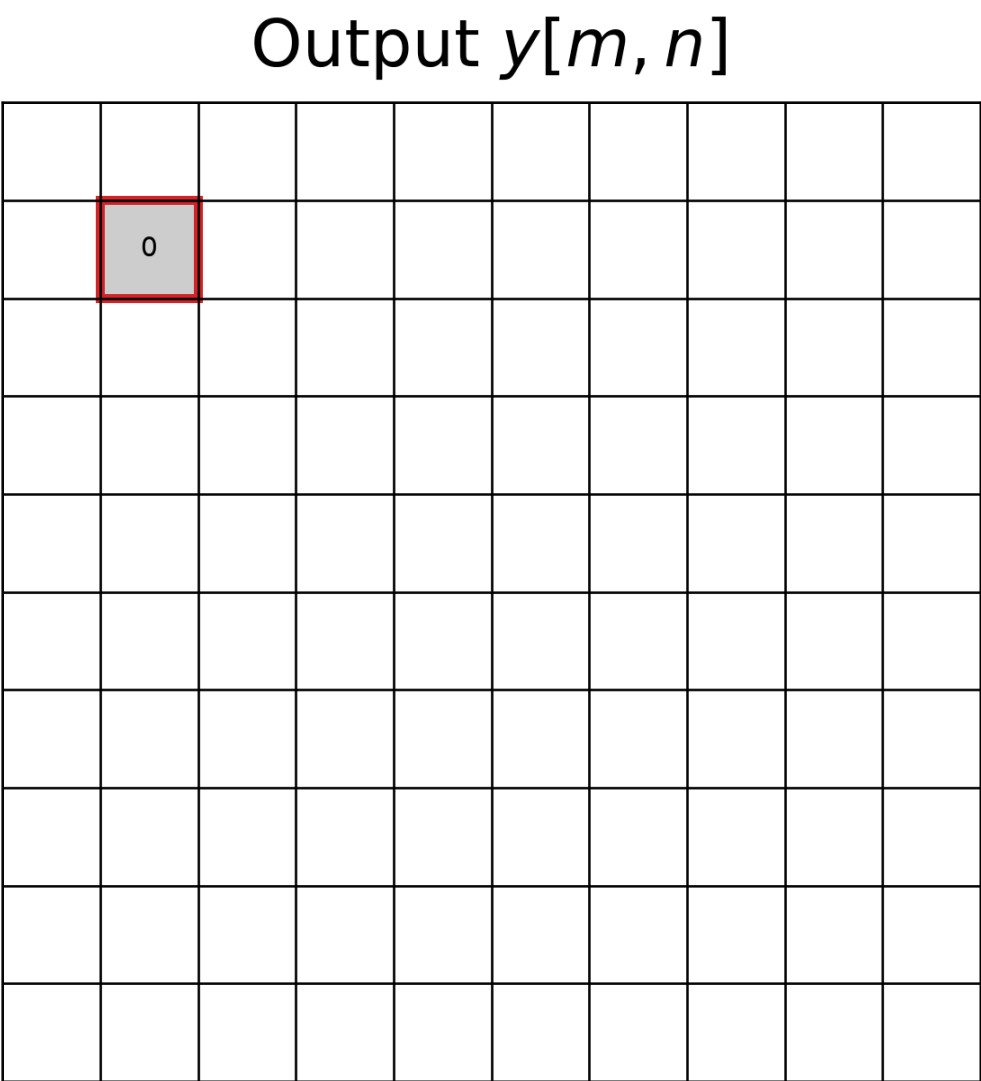
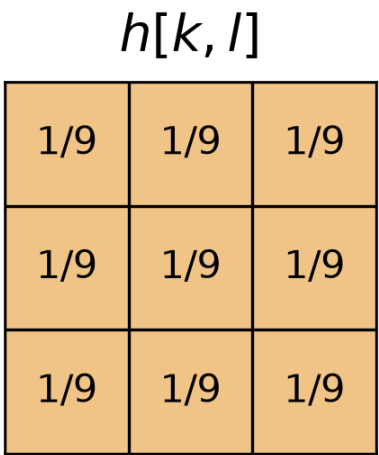
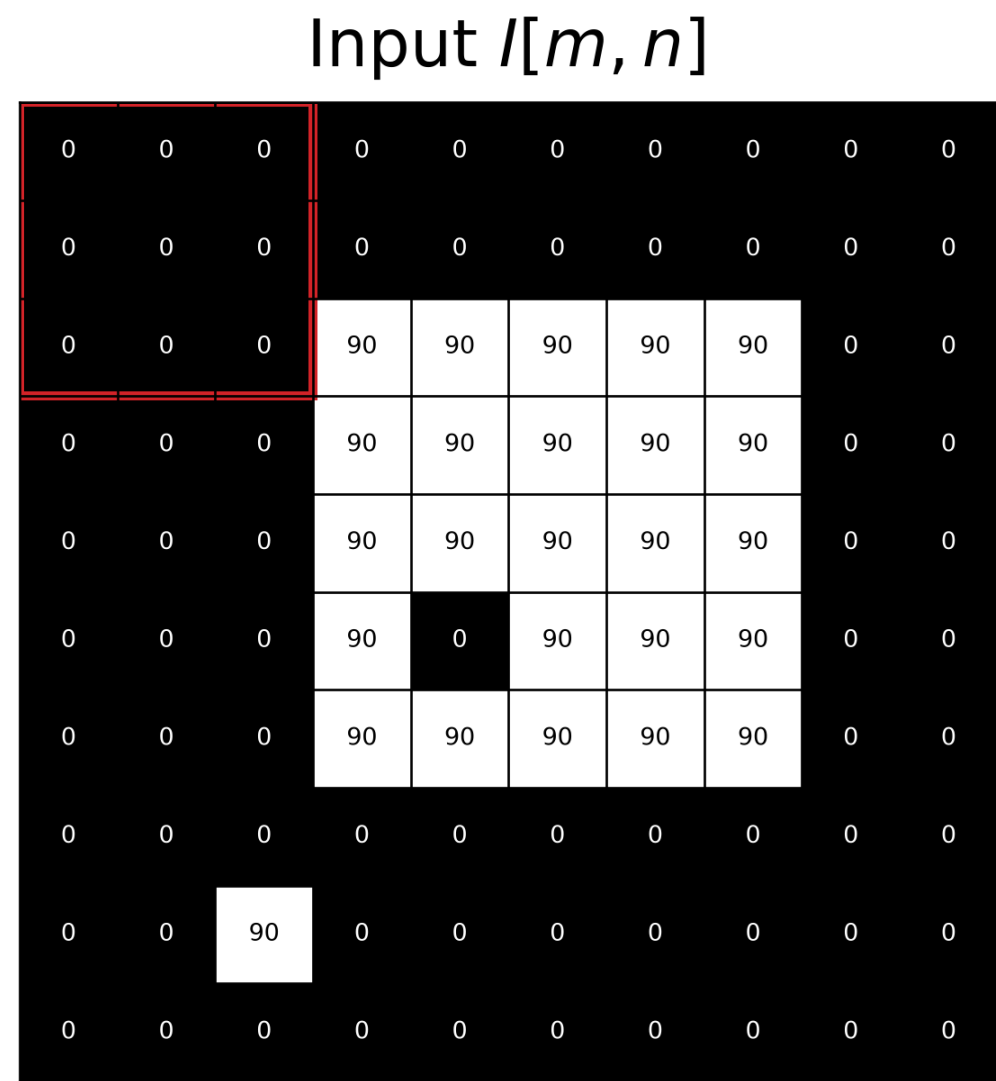
# Example: Box Filter Kernel

- The **box filter** replaces each pixel with the average of its neighborhood.
- The size of the neighborhood is determined by the size of the filter kernel
- For example, a  $3 \times 3$  box filter has the kernel:

$$h[k, l] = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$



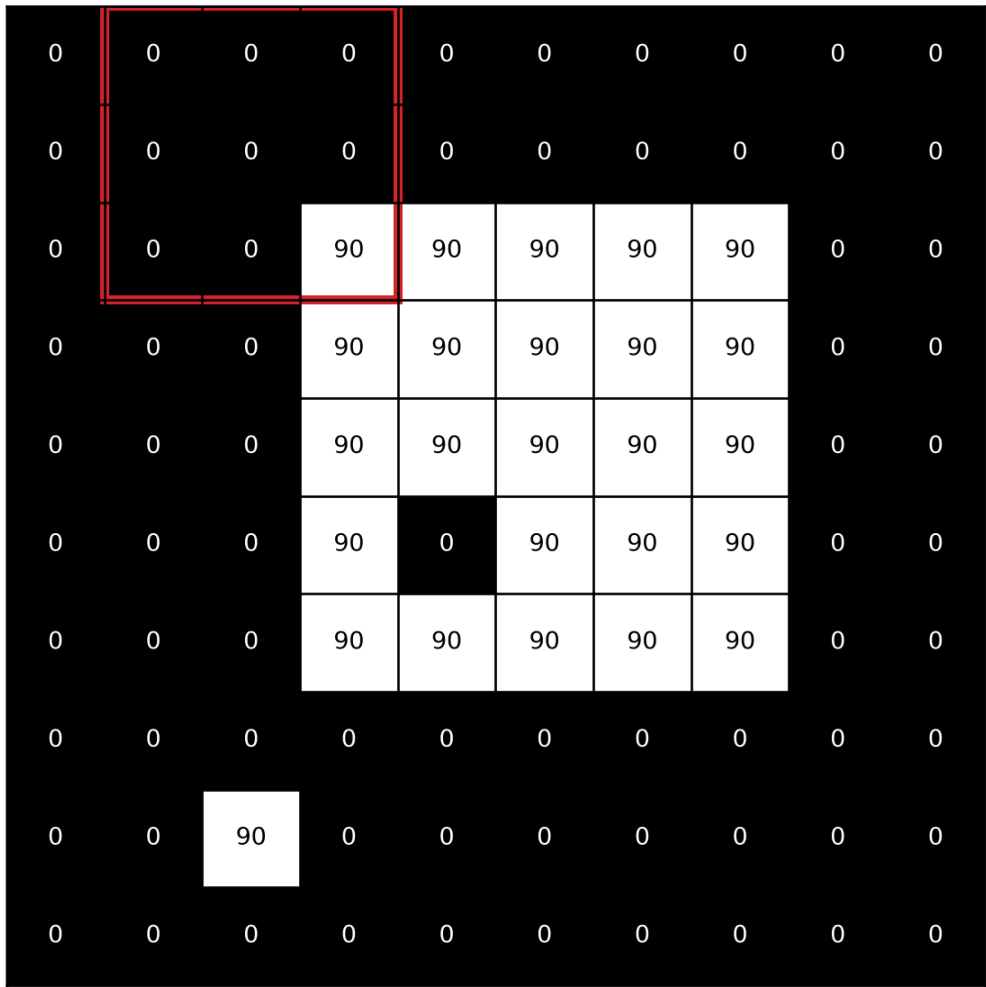
# Box Filter: First Position



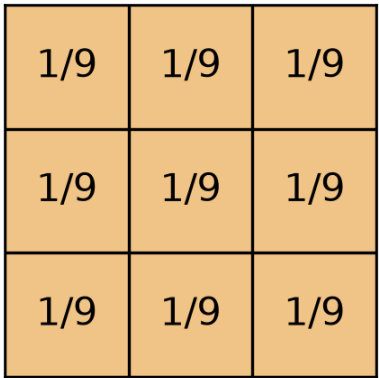
$$y[1, 1] = \sum_{k=-1}^1 \sum_{l=-1}^1 h[k, l] I[1 + k, 1 + l] = 0$$

# Box Filter: Move One Pixel Right

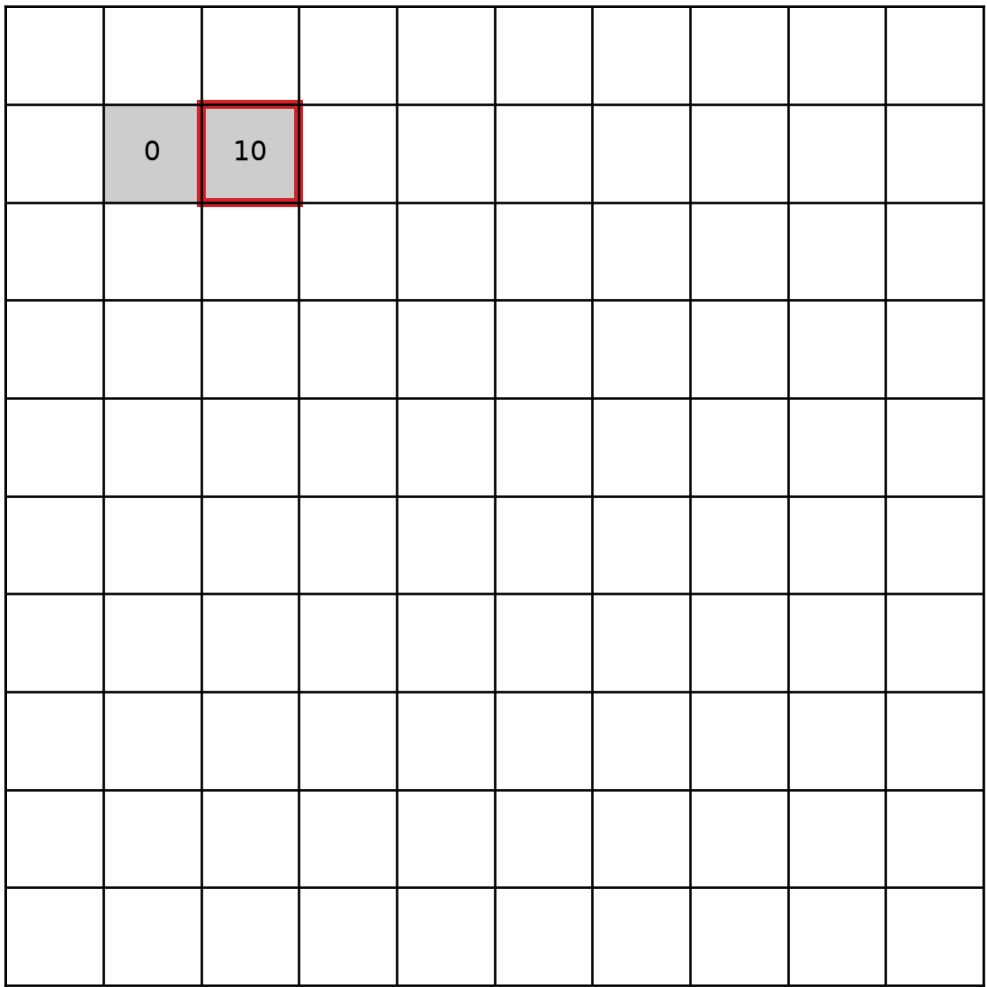
Input  $I[m, n]$



$h[k, l]$

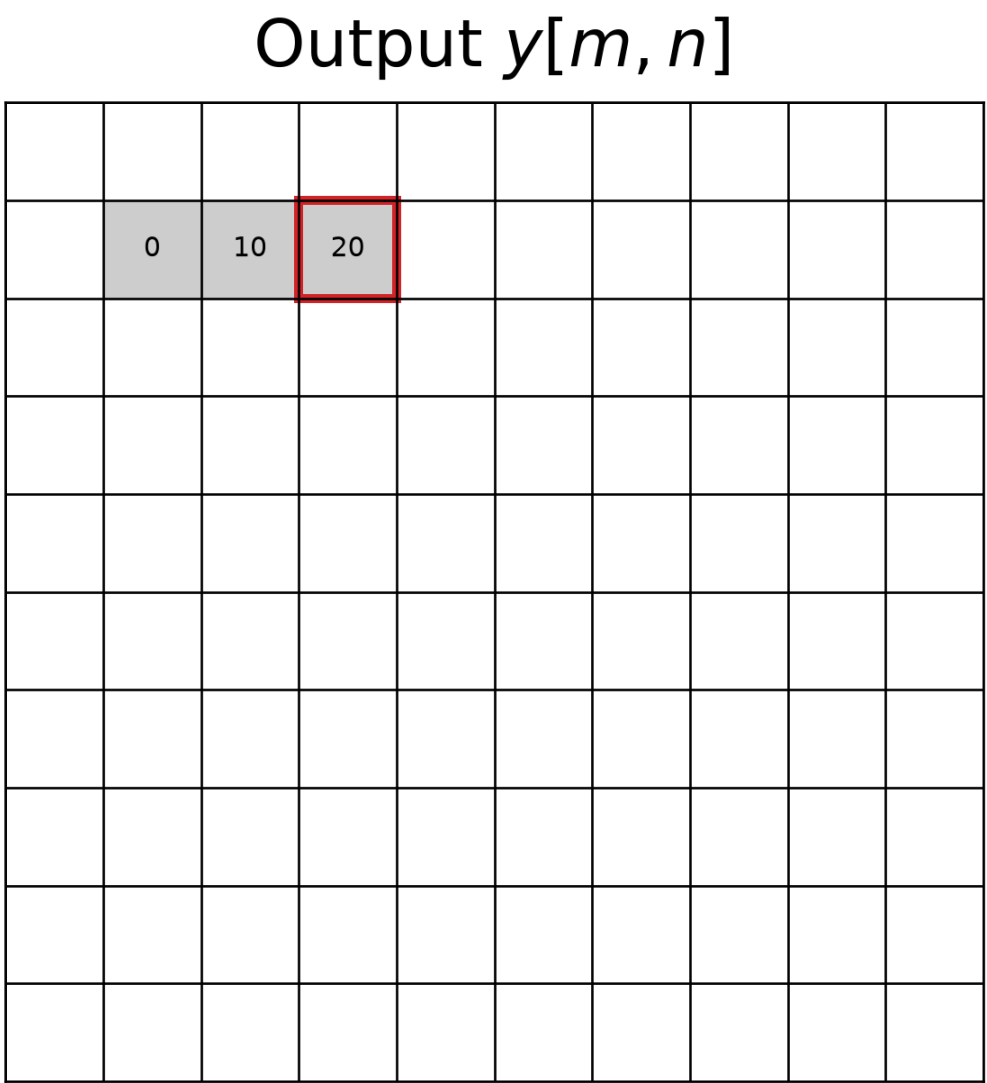
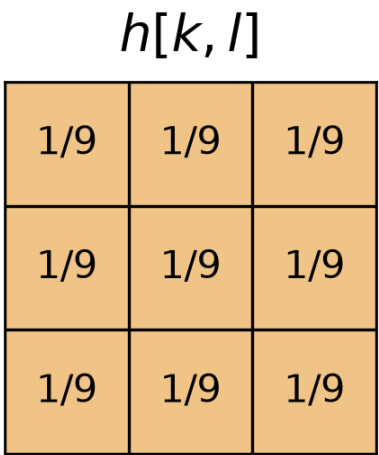
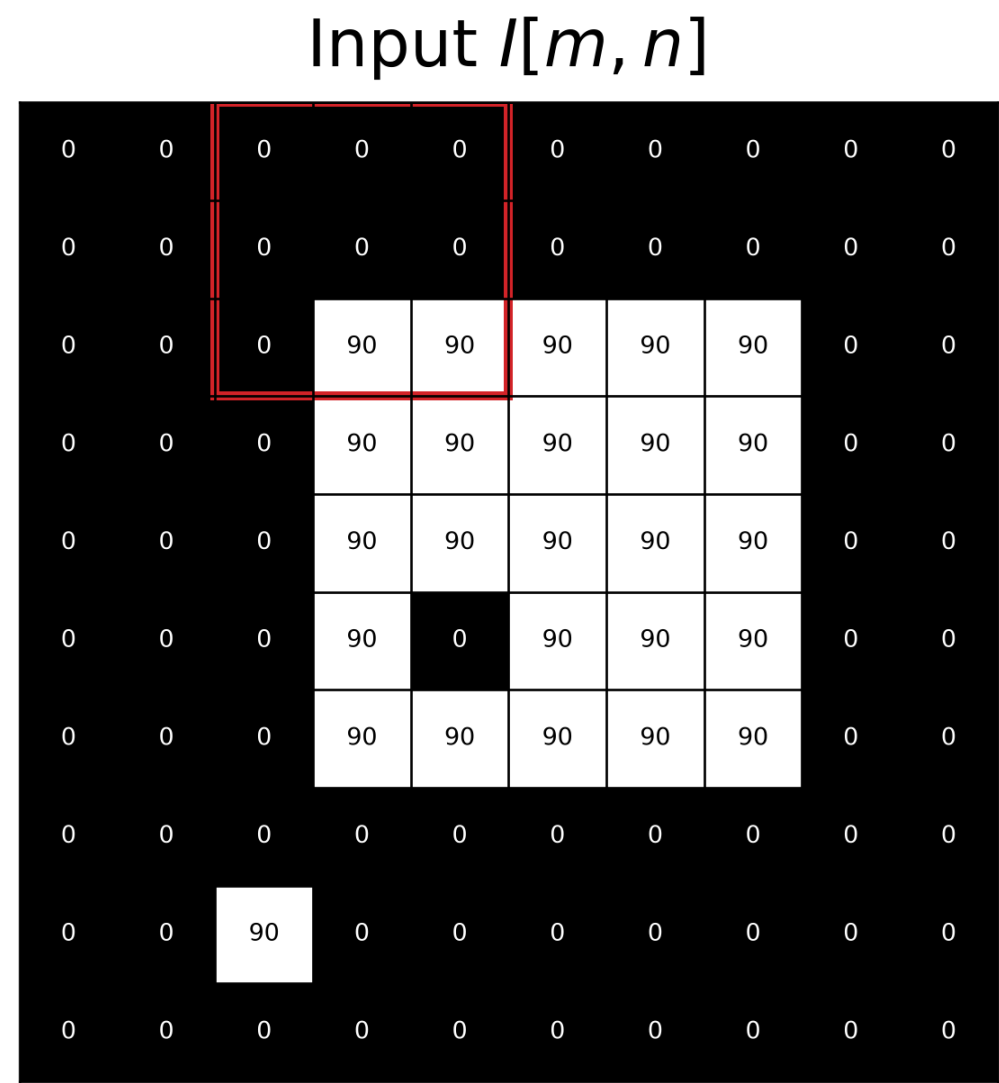


Output  $y[m, n]$



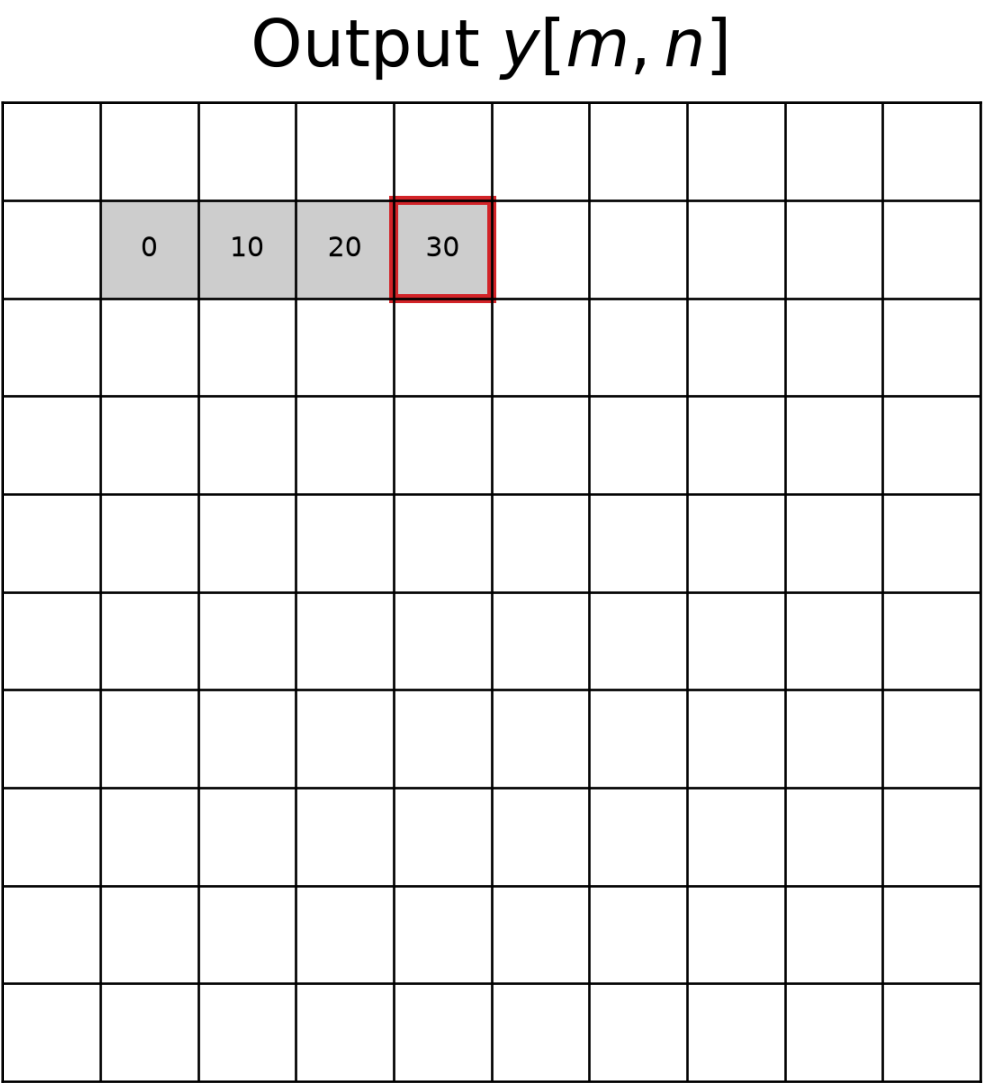
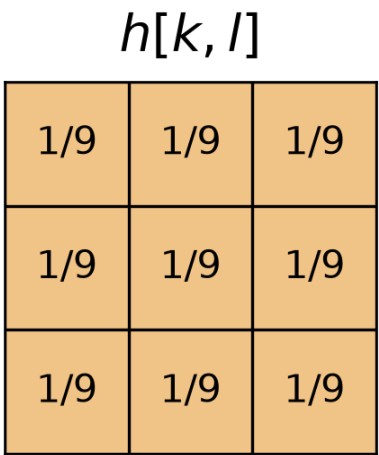
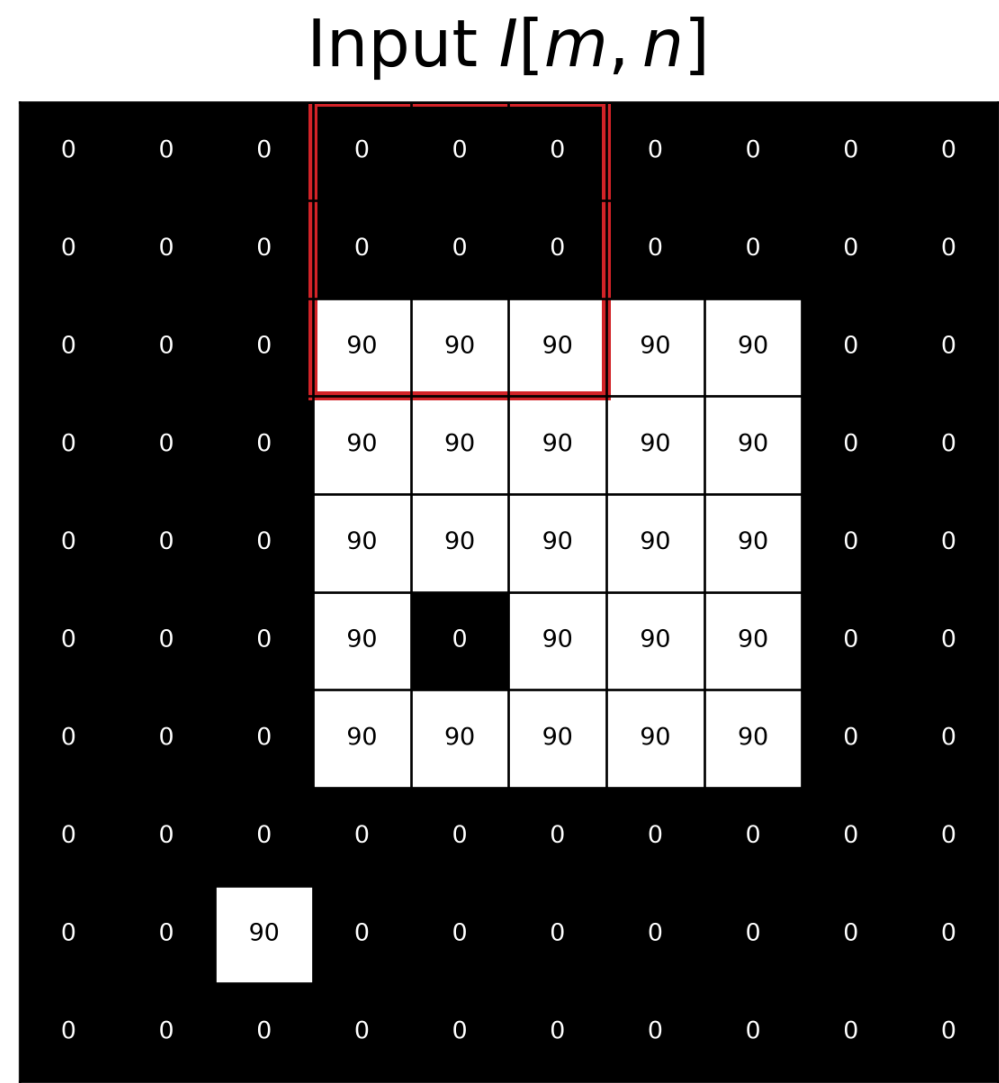
$$y[1, 2] = \sum_{k=-1}^1 \sum_{l=-1}^1 h[k, l] I[1 + k, 2 + l] = 10$$

# Box Filter: Continue the Sweep



$$y[1, 3] = \sum_{k=-1}^1 \sum_{l=-1}^1 h[k, l] I[1 + k, 3 + l] = 20$$

# Box Filter: Continue the Sweep



$$y[1, 4] = \sum_{k=-1}^1 \sum_{l=-1}^1 h[k, l] I[1 + k, 4 + l] = 30$$

# Box Filter: Continue the Sweep

Input  $I[m, n]$

|   |   |    |    |    |    |    |    |   |   |
|---|---|----|----|----|----|----|----|---|---|
| 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 |
| 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 |
| 0 | 0 | 0  | 90 | 90 | 90 | 90 | 90 | 0 | 0 |
| 0 | 0 | 0  | 90 | 90 | 90 | 90 | 90 | 0 | 0 |
| 0 | 0 | 0  | 90 | 90 | 90 | 90 | 90 | 0 | 0 |
| 0 | 0 | 0  | 90 | 0  | 90 | 90 | 90 | 0 | 0 |
| 0 | 0 | 0  | 90 | 90 | 90 | 90 | 90 | 0 | 0 |
| 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 |
| 0 | 0 | 90 | 0  | 0  | 0  | 0  | 0  | 0 | 0 |
| 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 |

$h[k, l]$

|     |     |     |
|-----|-----|-----|
| 1/9 | 1/9 | 1/9 |
| 1/9 | 1/9 | 1/9 |
| 1/9 | 1/9 | 1/9 |

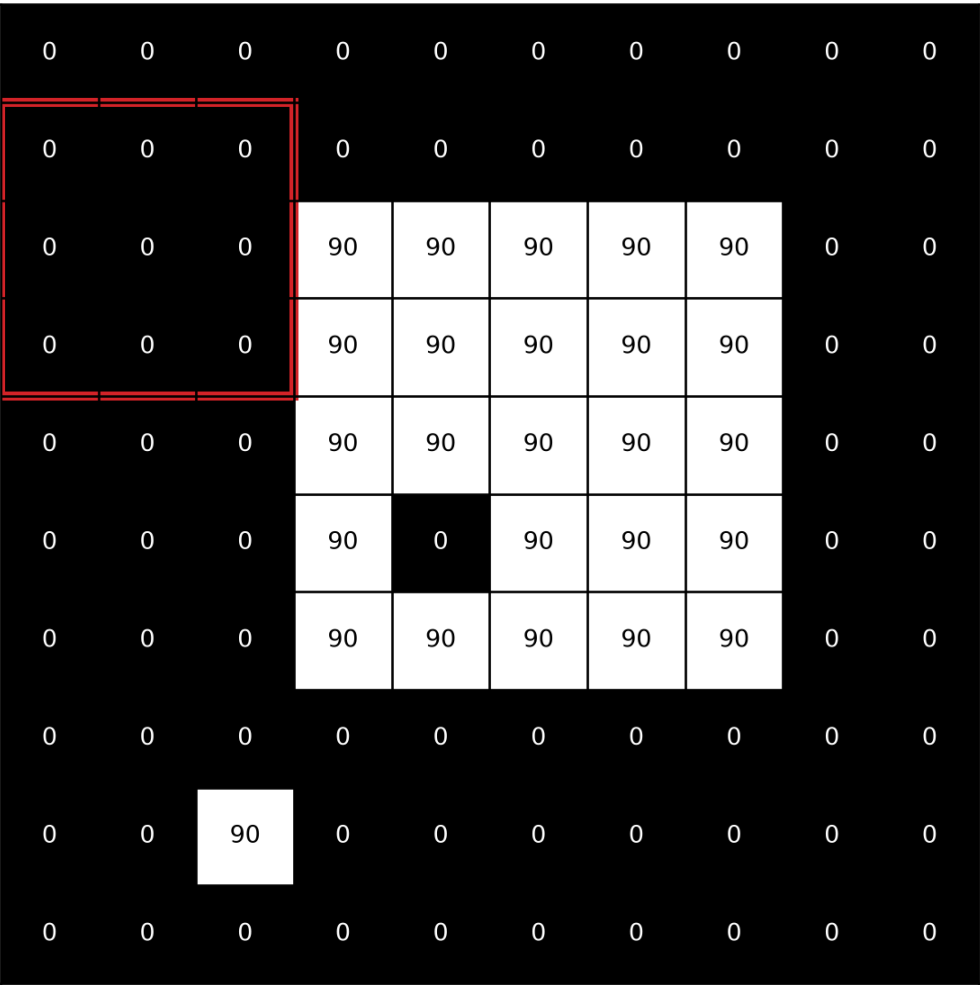
Output  $y[m, n]$

|  |   |    |    |    |    |    |    |    |  |
|--|---|----|----|----|----|----|----|----|--|
|  |   |    |    |    |    |    |    |    |  |
|  | 0 | 10 | 20 | 30 | 30 | 30 | 20 | 10 |  |
|  |   |    |    |    |    |    |    |    |  |
|  |   |    |    |    |    |    |    |    |  |
|  |   |    |    |    |    |    |    |    |  |
|  |   |    |    |    |    |    |    |    |  |
|  |   |    |    |    |    |    |    |    |  |
|  |   |    |    |    |    |    |    |    |  |
|  |   |    |    |    |    |    |    |    |  |
|  |   |    |    |    |    |    |    |    |  |

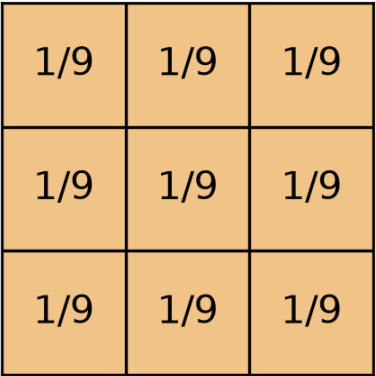
$$y[1, 8] = \sum_{k=-1}^1 \sum_{l=-1}^1 h[k, l] I[1 + k, 8 + l] = 10$$

# Box Filter: Continue Across the Column

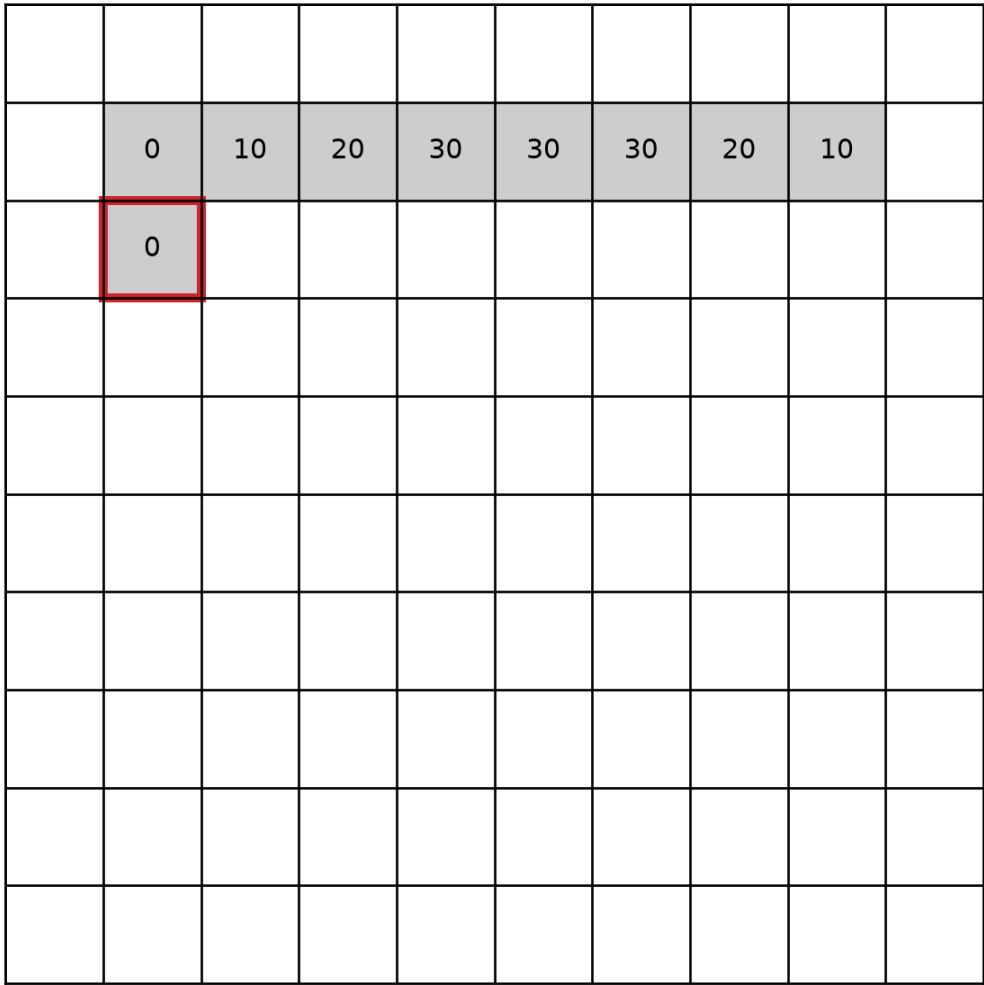
Input  $I[m, n]$



$h[k, l]$

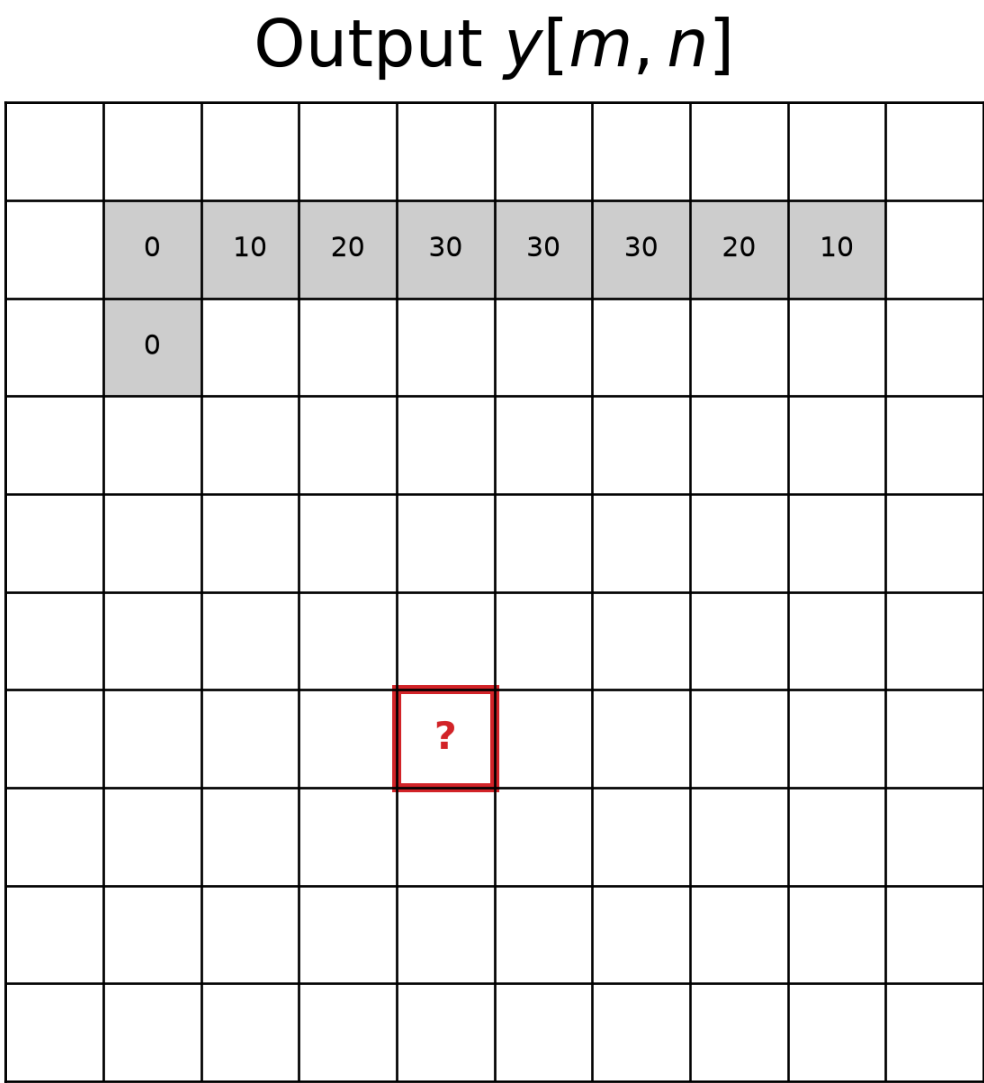
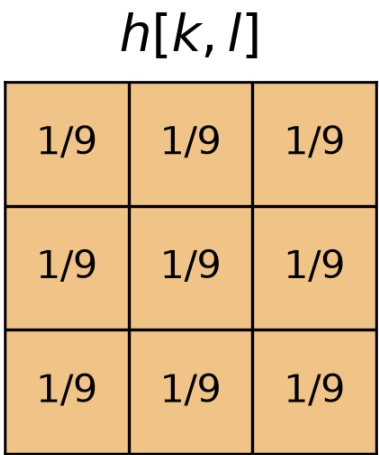
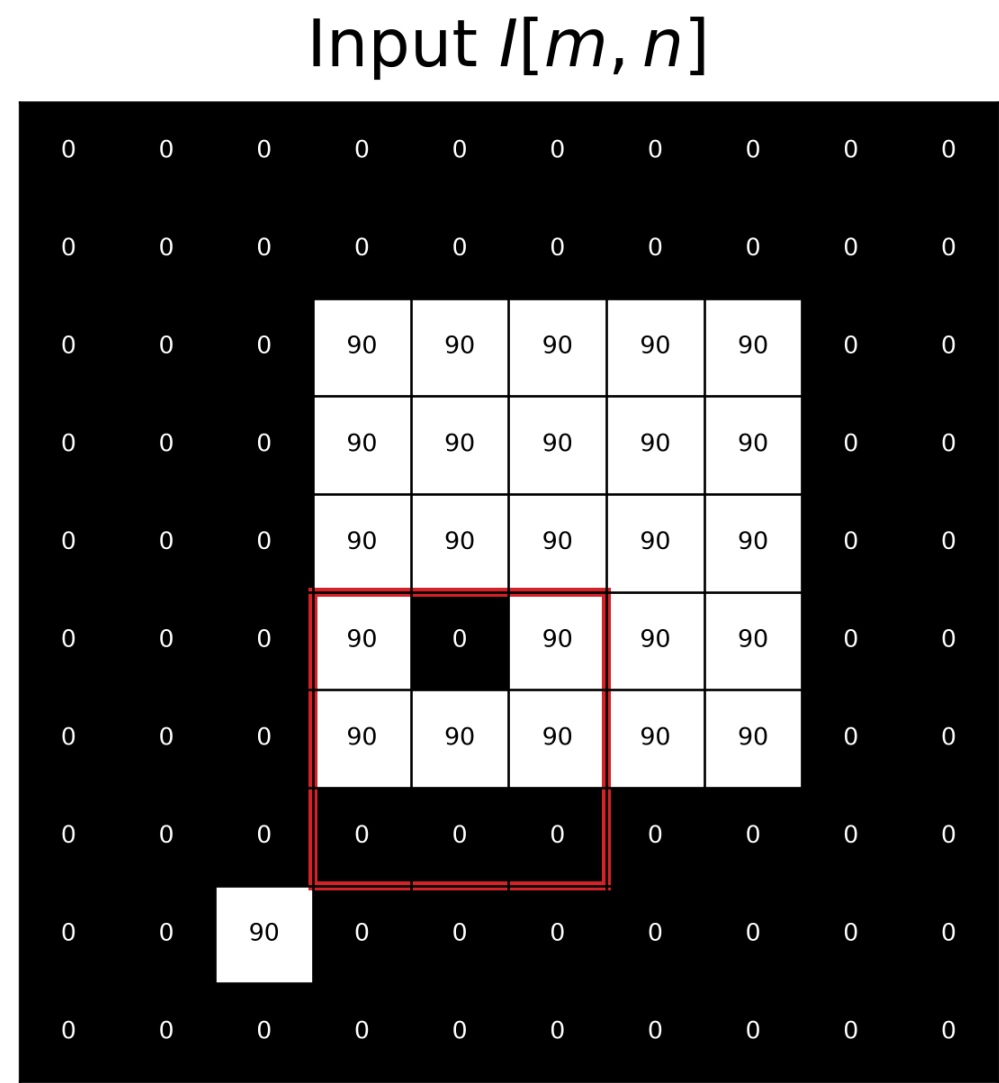


Output  $y[m, n]$



$$y[2, 1] = \sum_{k=-1}^1 \sum_{l=-1}^1 h[k, l] I[2 + k, 1 + l] = 0$$

# Box Filter: Try This Position



What is  $y[6, 4]$ ?

Add the nine pixels under the red window, then divide by nine.



# Box Filter: Another Position

Input  $I[m, n]$

|   |   |    |    |    |    |    |    |   |   |
|---|---|----|----|----|----|----|----|---|---|
| 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 |
| 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 |
| 0 | 0 | 0  | 90 | 90 | 90 | 90 | 90 | 0 | 0 |
| 0 | 0 | 0  | 90 | 90 | 90 | 90 | 90 | 0 | 0 |
| 0 | 0 | 0  | 90 | 90 | 90 | 90 | 90 | 0 | 0 |
| 0 | 0 | 0  | 90 | 0  | 90 | 90 | 90 | 0 | 0 |
| 0 | 0 | 0  | 90 | 90 | 90 | 90 | 90 | 0 | 0 |
| 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 |
| 0 | 0 | 90 | 0  | 0  | 0  | 0  | 0  | 0 | 0 |
| 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 |

$h[k, l]$

|     |     |     |
|-----|-----|-----|
| 1/9 | 1/9 | 1/9 |
| 1/9 | 1/9 | 1/9 |
| 1/9 | 1/9 | 1/9 |

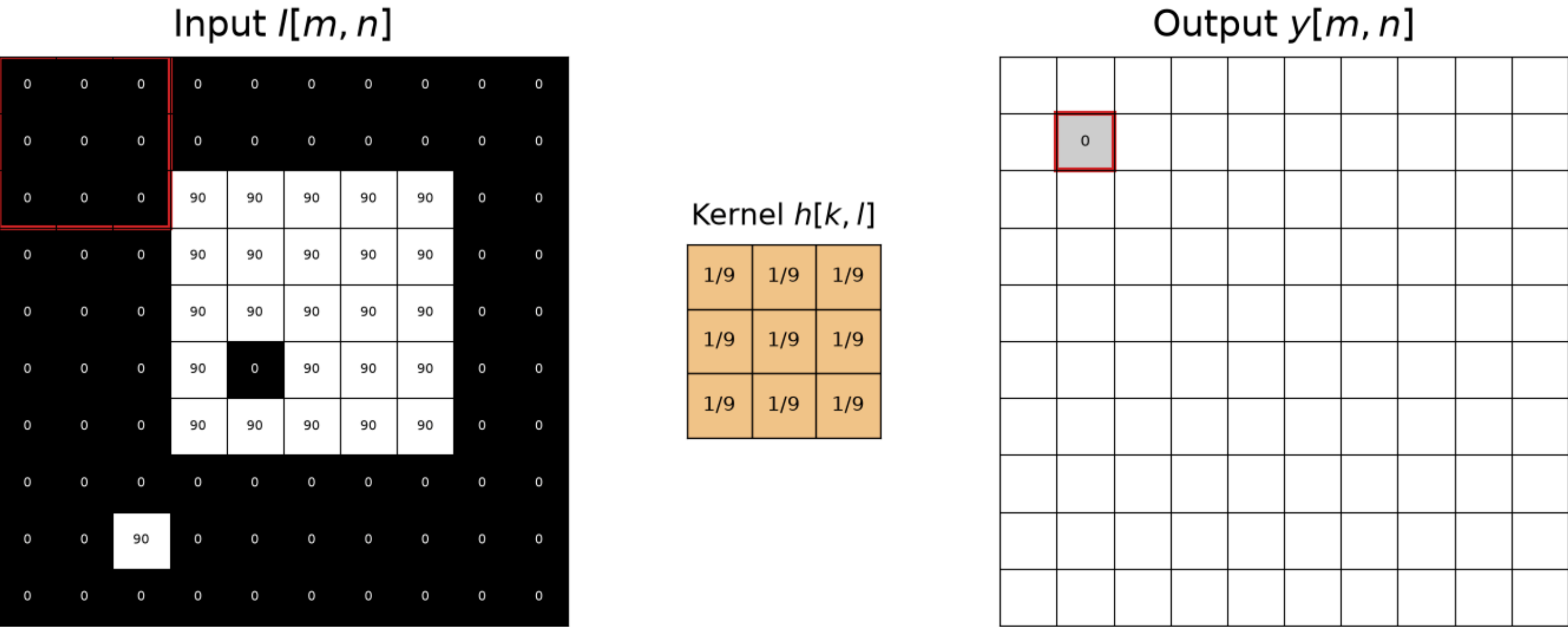
Output  $y[m, n]$

|  |   |    |    |    |    |    |    |    |  |
|--|---|----|----|----|----|----|----|----|--|
|  |   |    |    |    |    |    |    |    |  |
|  | 0 | 10 | 20 | 30 | 30 | 30 | 20 | 10 |  |
|  | 0 |    |    |    |    |    |    |    |  |
|  |   |    |    |    |    |    |    |    |  |
|  |   |    |    |    |    | ?  |    |    |  |
|  |   |    |    |    |    |    |    |    |  |
|  |   |    |    | 50 |    |    |    |    |  |
|  |   |    |    |    |    |    |    |    |  |
|  |   |    |    |    |    |    |    |    |  |
|  |   |    |    |    |    |    |    |    |  |

What is  $y[4, 6]$ ?



# Box Filter: Complete Output



Position  $(m, n) = (1, 1)$ :  $y[1, 1] = 0$

# Box Filter: Complete Output

Filtered output  $y[m, n]$

|  |    |    |    |    |    |    |    |    |  |
|--|----|----|----|----|----|----|----|----|--|
|  |    |    |    |    |    |    |    |    |  |
|  | 0  | 10 | 20 | 30 | 30 | 30 | 20 | 10 |  |
|  | 0  | 20 | 40 | 60 | 60 | 60 | 40 | 20 |  |
|  | 0  | 30 | 60 | 90 | 90 | 90 | 60 | 30 |  |
|  | 0  | 30 | 50 | 80 | 80 | 90 | 60 | 30 |  |
|  | 0  | 30 | 50 | 80 | 80 | 90 | 60 | 30 |  |
|  | 0  | 20 | 30 | 50 | 50 | 60 | 40 | 20 |  |
|  | 10 | 20 | 30 | 30 | 30 | 30 | 20 | 10 |  |
|  | 10 | 10 | 10 | 0  | 0  | 0  | 0  | 0  |  |
|  |    |    |    |    |    |    |    |    |  |

$$y[m, n] = \sum_{k=-1}^1 \sum_{l=-1}^1 h[k, l] I[m + k, n + l]$$

# Box Filter: What Does It Do?

- Replaces each pixel by the average of its local neighborhood.
- Smooths small intensity variations and noise.
- Suppresses sharp edges and fine details.
- Why does the kernel sum to one?

$$h[k, l] = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

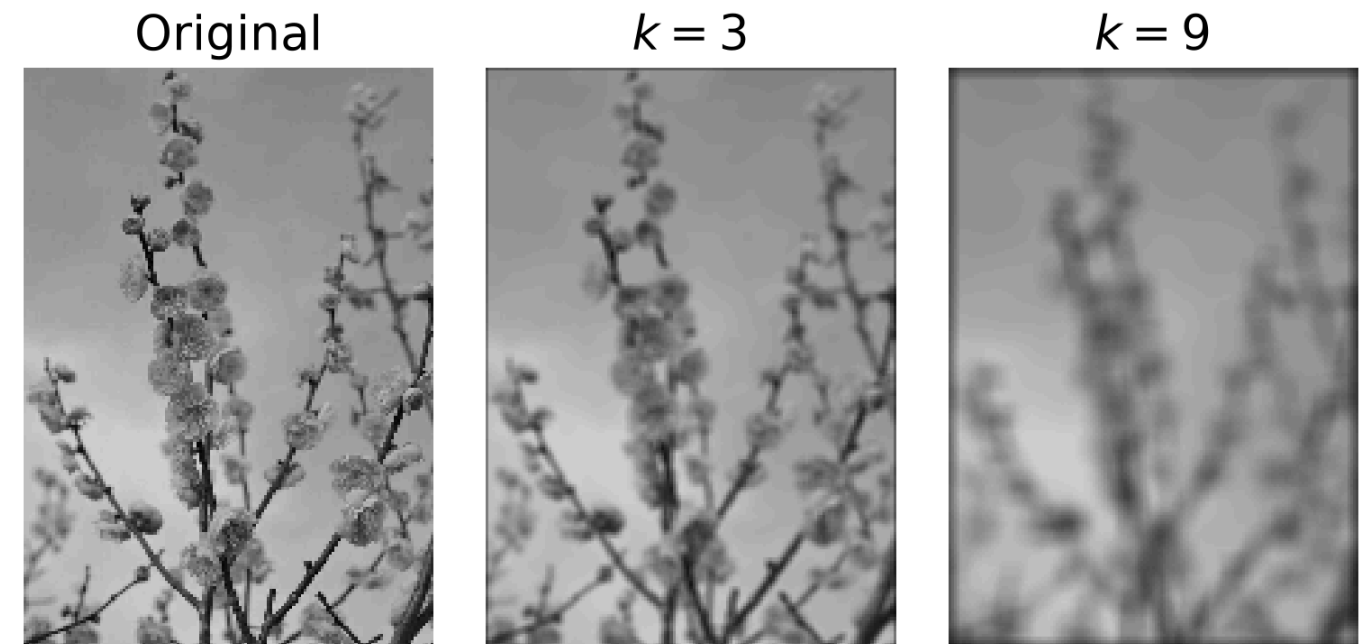
## ! Important

The coefficients sum to one, so the filter preserves constant image regions.

# Smoothing with a Box Filter

- For a  $k \times k$  box filter,  $h[k, k] = 1/k^2$ .
- A larger  $k$  produces stronger smoothing

```
1 import cv2
2
3 img_path = Path("assets/03_filtering/bloom_o
4 img_col = cv2.imread(str(img_path))
5
6 img_gray = cv2.cvtColor(img_col, cv2.COLOR_B
7 img_flt = img_gray.astype(np.float32) / 255.0
8 img = cv2.resize(img_flt, None, fx=0.2, fy=0
9
10 box3 = cv2.blur(img, (3, 3), borderType=cv2.B
11 box9 = cv2.blur(img, (9, 9), borderType=cv2.B
12
13 fig, axes = plt.subplots(1, 3, figsize=(6.5,
14 for ax, result, title in zip(
15     axes, [img, box3, box9], ["Original", "$
16     ax.imshow(result, cmap="gray", vmin=0, vi
17     ax.set_title(title) + ax.axis('off')
```

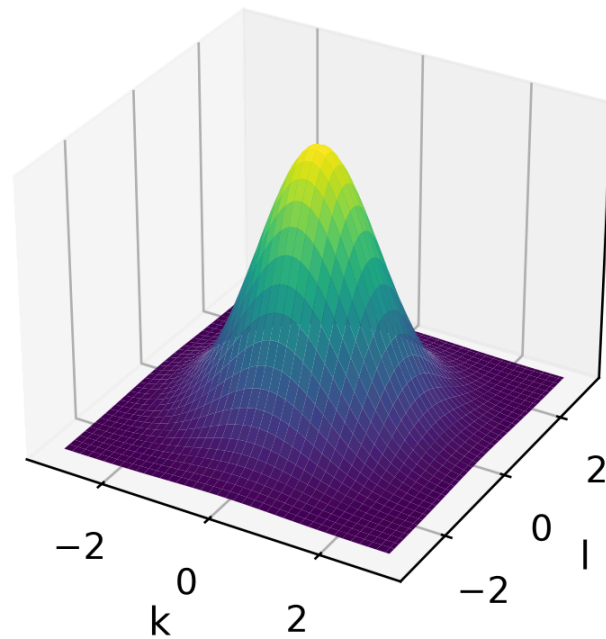


# Gaussian Filter Kernel

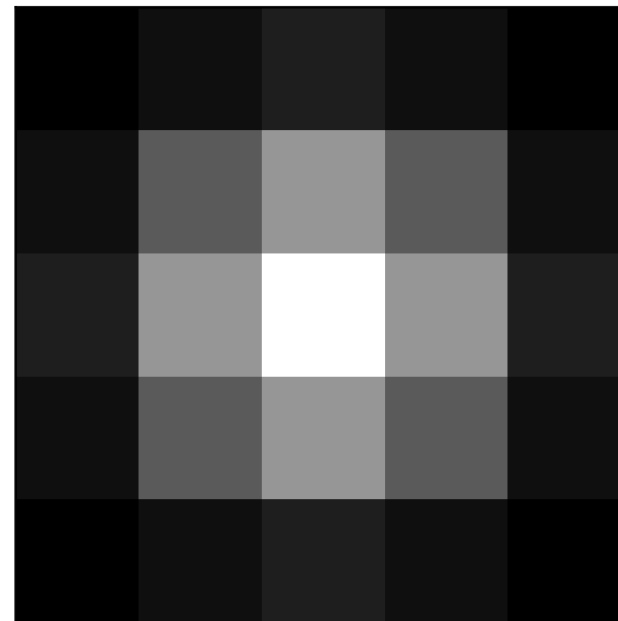
A Gaussian kernel gives the largest weight to the center pixel and smoothly decreases with distance:

$$h_{\sigma}[k, l] = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{k^2 + l^2}{2\sigma^2}\right).$$

Continuous Gaussian



5 × 5 sampled kernel

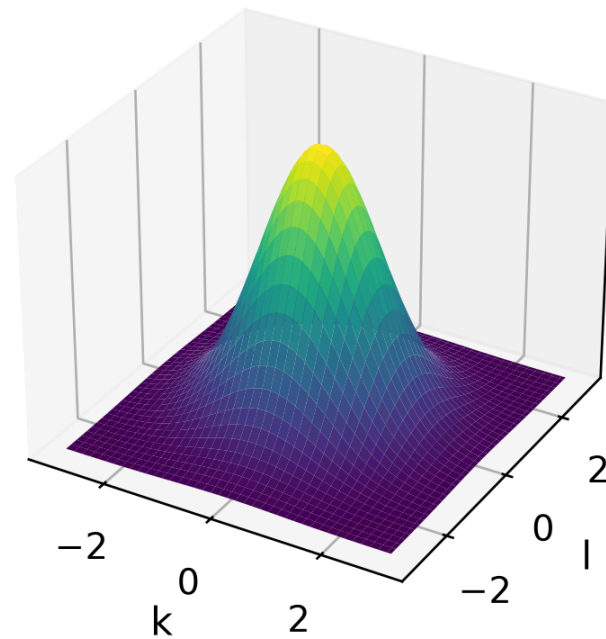


5 × 5 kernel,  $\sigma = 1$

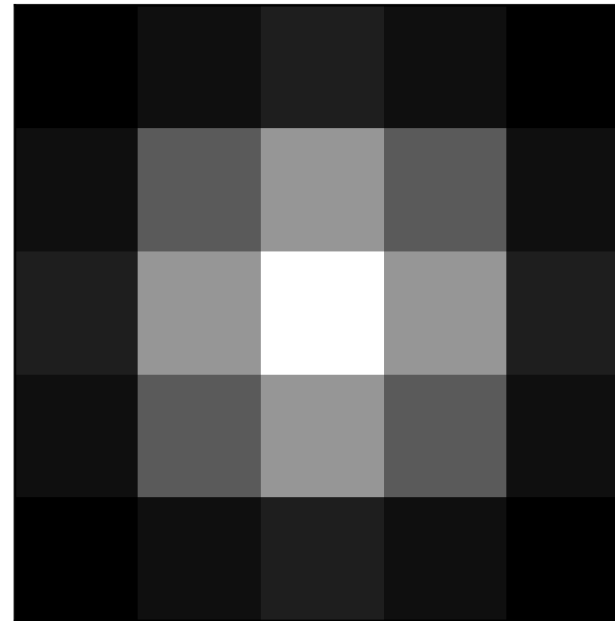
|       |       |       |       |       |
|-------|-------|-------|-------|-------|
| 0.003 | 0.013 | 0.022 | 0.013 | 0.003 |
| 0.013 | 0.060 | 0.098 | 0.060 | 0.013 |
| 0.022 | 0.098 | 0.162 | 0.098 | 0.022 |
| 0.013 | 0.060 | 0.098 | 0.060 | 0.013 |
| 0.003 | 0.013 | 0.022 | 0.013 | 0.003 |

# Gaussian Filter Kernel

Continuous Gaussian



5 × 5 sampled kernel



5 × 5 kernel,  $\sigma = 1$

|       |       |       |       |       |
|-------|-------|-------|-------|-------|
| 0.003 | 0.013 | 0.022 | 0.013 | 0.003 |
| 0.013 | 0.060 | 0.098 | 0.060 | 0.013 |
| 0.022 | 0.098 | 0.162 | 0.098 | 0.022 |
| 0.013 | 0.060 | 0.098 | 0.060 | 0.013 |
| 0.003 | 0.013 | 0.022 | 0.013 | 0.003 |

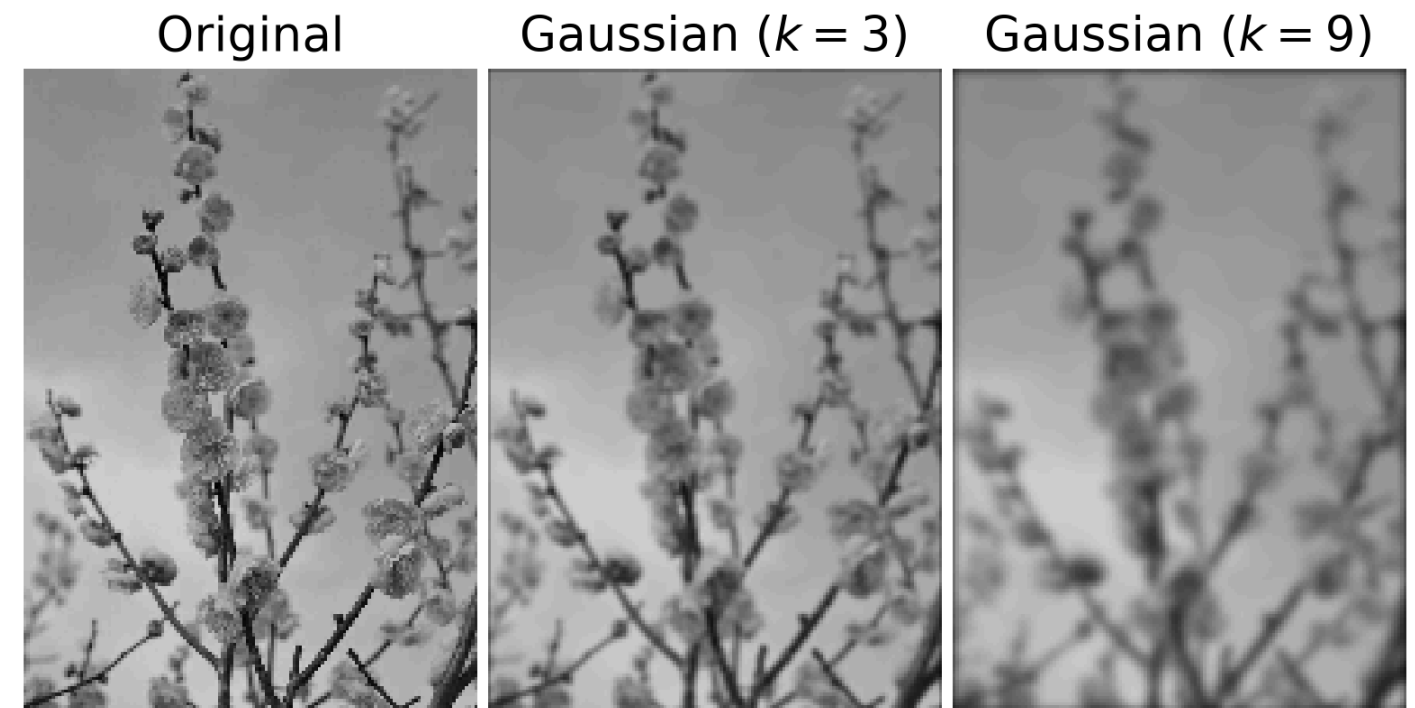
## *i* Note

- Gaussian kernel weights sum to one (aside from negligible floating-point errors) – **why?**
- The continuous Gaussian integrates to one over the infinite plane. A truncated (5) sample does not automatically sum to one, so the explicit normalization line is necessary.
- Sampled kernel size should be large enough to include the meaningful support of the Gaussian—often several standard deviations on each side.

# Smoothing with a Gaussian Filter

- A Gaussian filter smooths noises.

```
1 import cv2
2
3 gau3 = cv2.GaussianBlur(img, (3, 3), sigmaX=0, sigmaY=0)
4 gau9 = cv2.GaussianBlur(img, (9, 9), sigmaX=0, sigmaY=0)
5
6 fig, axes = plt.subplots(1, 3, figsize=(6.5, 4.5))
7 for ax, im, title in zip(
8     axes, [img, gau3, gau9],
9     ["Original", r"Gaussian ($k=3$)", r"Gaussian ($k=9$)"]):
10     ax.imshow(im, cmap="gray", vmin=0, vmax=255)
11     ax.set_title(title)
12     ax.axis("off")
13 plt.tight_layout(pad=0.3)
14 plt.show()
```





# Box versus Gaussian Filter Smoothing

Original



Box filter ( $k = 9$ )



Gaussian filter ( $k = 9$ )



- The box filter weights every pixel in the window equally.
- The Gaussian filter changes weights smoothly, which produces fewer artifacts.



# Linear Filter Properties

## Linearity:

$$\text{filter}(I, h_1 + h_2) = \text{filter}(I, h_1) + \text{filter}(I, h_2)$$

$h_1$

|   |   |   |
|---|---|---|
| 1 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |

$+ \quad h_2$

|   |   |   |
|---|---|---|
| 0 | 1 | 0 |
| 1 | 1 | 1 |
| 0 | 1 | 0 |

$= \quad h_1 + h_2$

|   |   |   |
|---|---|---|
| 1 | 1 | 1 |
| 1 | 2 | 1 |
| 1 | 1 | 1 |

## Shift/translation invariance:

$$\text{filter}(I, \text{shift}(h)) = \text{shift}(\text{filter}(I, h))$$

$h$  at position  $(m, n)$

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 1 | 0 |
| 1 | 2 | 1 | 0 |
| 1 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

shift  
→

$h$  at position  $(m + 1, n + 1)$

|   |   |   |   |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 0 | 1 | 2 | 1 |
| 0 | 1 | 1 | 1 |

# Separable Filters

A 2D kernel  $h$  is **separable** if

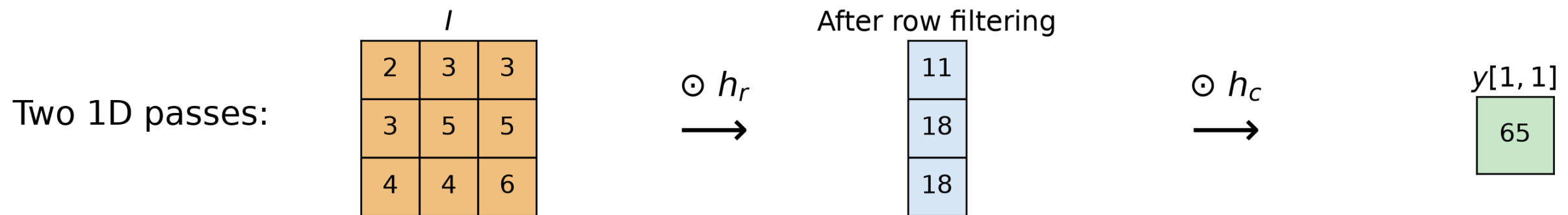
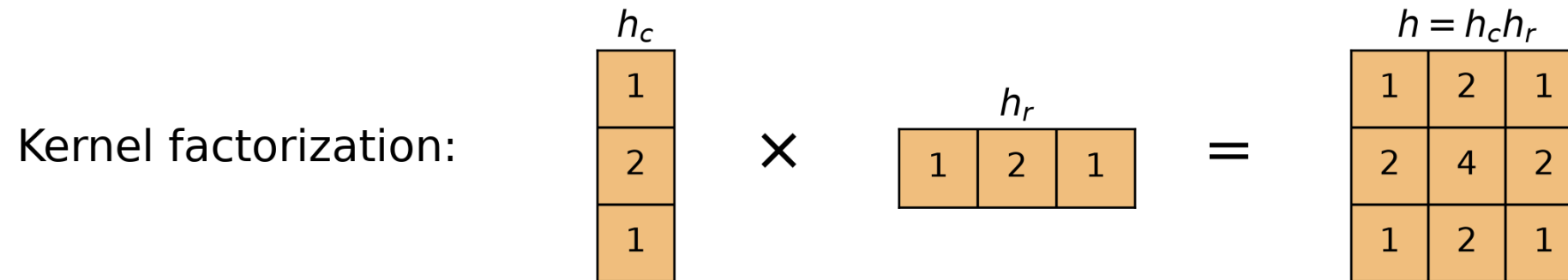
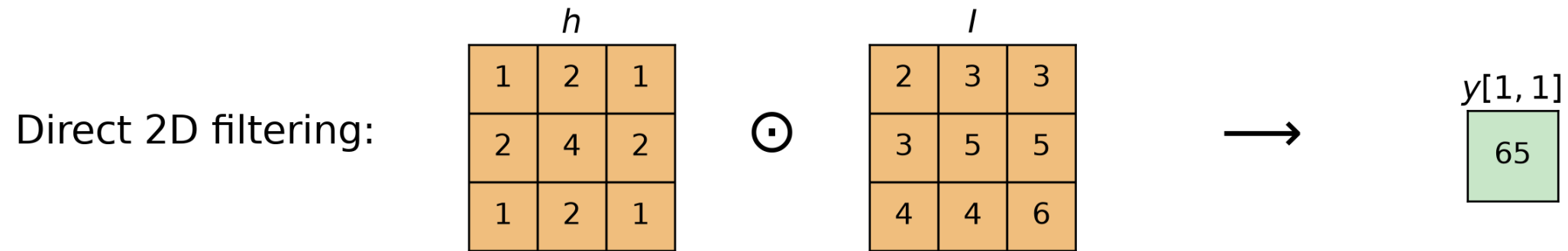
$$h = v u^T,$$

where  $v$  is a vertical 1D kernel and  $u^T$  is a horizontal 1D kernel.

For example,

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \frac{1}{4} [1 \quad 2 \quad 1].$$

# Separability Example



# Separability Reduces Computation

For an  $M \times N$  image and a  $P \times Q$  kernel:

| Method              | Approximate multiply-adds |
|---------------------|---------------------------|
| Full 2D filtering   | $MNPQ$                    |
| Separable filtering | $MN(P + Q)$               |

For a square  $K \times K$  kernel, the *per-pixel* cost drops from  $K^2$  to  $2K$ .

**Example:** A separable  $9 \times 9$  filter needs about  $81/18 = 4.5$  times fewer multiply-adds than direct 2D filtering.

# The Gaussian Filter Is Separable

Because

$$\frac{1}{2\pi\sigma^2} e\left(-\frac{x^2+y^2}{2\sigma^2}\right) = \left(\frac{1}{\sqrt{2\pi}\sigma} e\left(-\frac{x^2}{2\sigma^2}\right)\right) \left(\frac{1}{\sqrt{2\pi}\sigma} e\left(-\frac{y^2}{2\sigma^2}\right)\right) \implies$$
$$G_\sigma(x, y) = G_\sigma(x) G_\sigma(y),$$

we can:

1. Filter every row with a 1D Gaussian.
  2. Filter every column with the same 1D Gaussian.
- Repeated filtering with smaller Gaussian kernels can produce the same result as filtering once with a larger Gaussian kernel.
  - Applying a Gaussian filter with standard deviation  $\sigma$  twice is equivalent to applying one Gaussian filter with  $\sigma_{\text{combined}} = \sqrt{2} \sigma$ .
  - This produces the same result with substantially fewer computations.

# Filtering: Correlation and Convolution

## 2D Correlation:

The kernel is applied without changing its orientation:

$$y[m, n] = \sum_{k, l} h[k, l] I[m+k, n+l].$$

OpenCV's `filter2D` computes correlation:

```
1 y_corr = cv2.filter2D(  
2     I, -1, h, borderType=cv2.BORDER_CONSTANT
```

## 2D Convolution:

The kernel indices are reversed:

$$y[m, n] = \sum_{k, l} h[k, l] I[m-k, n-l].$$

For convolution, **rotate the kernel by 180° first**:

```
1 h_rotated = cv2.flip(h, -1)  
2  
3 y_conv = cv2.filter2D(  
4     I, -1, h_rotated, borderType=cv2.BORDER_
```

- Convolution is **equivalent to correlation with a 180° rotated kernel**:

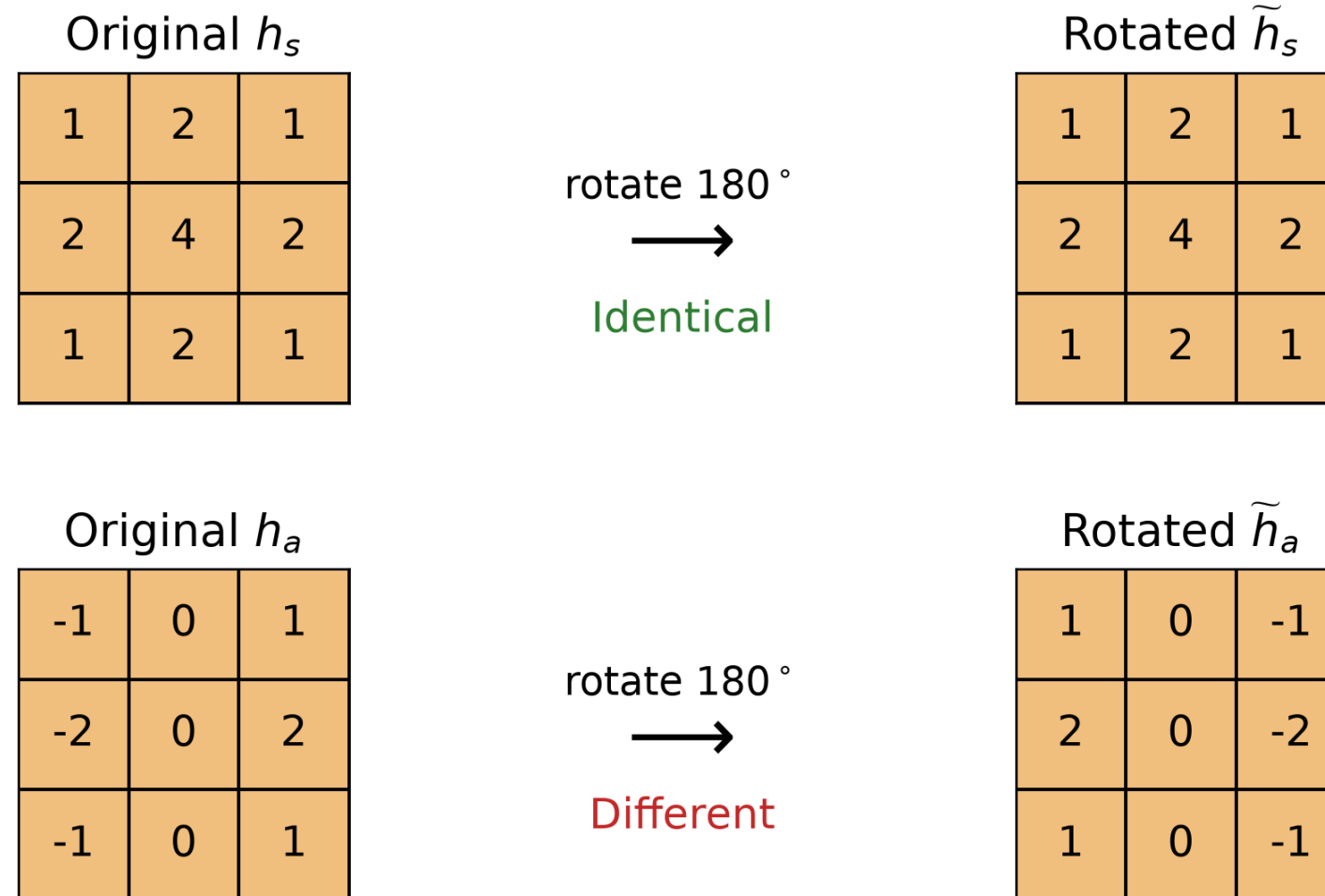
$$\tilde{h}[k, l] = h[-k, -l].$$

- Correlation and convolution are **identical** when the kernel has **180° rotational symmetry**:

$$h[k, l] = h[-k, -l].$$

# Correlation and Convolution Differ for Asymmetric Kernels

A kernel rotated by  $180^\circ$ :  $\tilde{h}[k, l] = h[-k, -l]$ .



- Since  $h_s = \tilde{h}_s$ , correlation and convolution produce the same result.
- But  $h_a \neq \tilde{h}_a$ , so correlation and convolution produce different responses.



# Convolution (\*) Properties

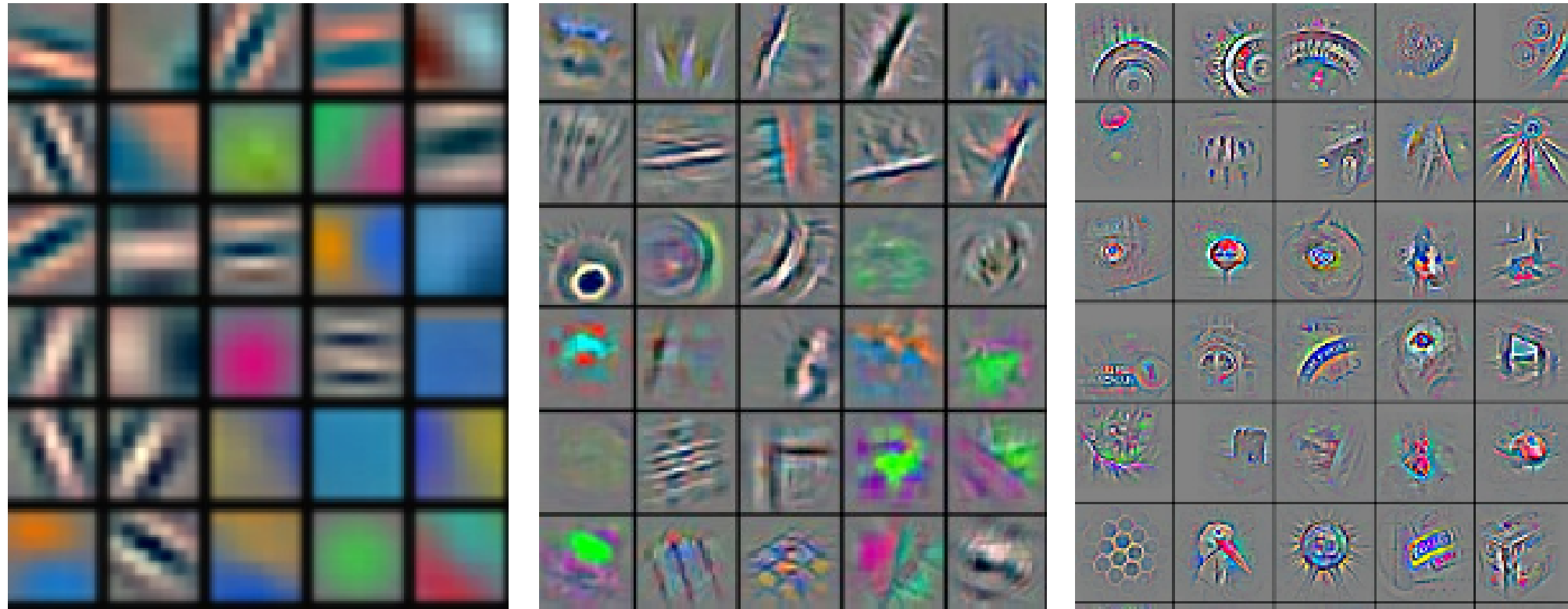
- **Commutative:**  $I * h = h * I$ 
  - Mathematically, the input and kernel can be interchanged.
- **Associative:**  $(I * h_1) * h_2 = I * (h_1 * h_2)$ 
  - A sequence of filters can be combined into a single kernel.
  - This can reduce the number of filtering operations.
  - Note that **correlation is not associative** in general.
- **Distributive:**  $I * (h_1 + h_2) = I * h_1 + I * h_2$
- **Scalar factor out:**  $(\alpha I) * h = \alpha(I * h)$
- **Identity:**  $I * \delta = I$ , where  $\delta$  is the unit impulse
  - The identity kernel is the unit impulse:
$$\delta[m, n] = \begin{cases} 1, & m = 0, n = 0, \\ 0, & \text{otherwise.} \end{cases}$$
  - Convolving with the unit impulse leaves the image unchanged

## ⚠ Important

- Finite images and boundary rules can make practical implementations appear to violate these identities near the edges.
- Convolution filters are both linear and shift-invariant – **why?**
- Every liner filter can be represented as convolution with some kernel – **why?**

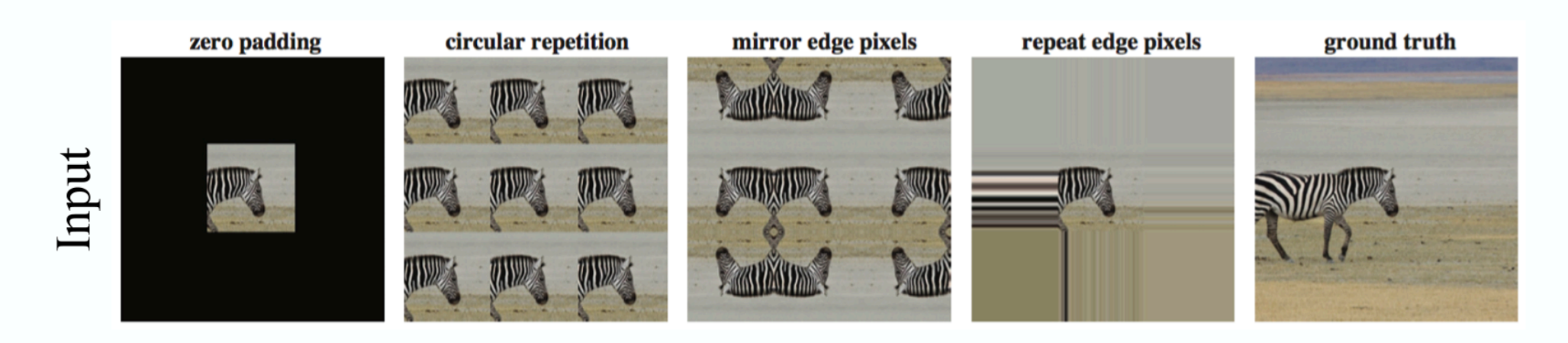
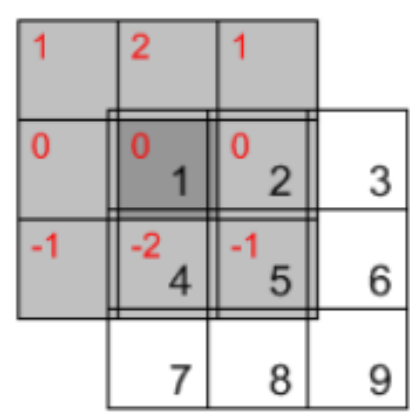


# “Convolution” Layers Usually Compute Correlation



- Convolutional neural networks learn their kernel weights from data rather than using predefined filters.
- Despite the name, most deep-learning libraries apply learned kernels without rotating them, which is, correlation.
- This does not reduce the network’s expressive power since the kernel is learned, and the network can learn either a kernel or its rotated version.

# Boundary Handling



Near an image boundary, part of the kernel falls outside the available data. Need to extrapolate (padding) to compute the output edge pixels.

| Strategy        | Assumption outside the image         | Common consequence                 |
|-----------------|--------------------------------------|------------------------------------|
| Constant / zero | Missing pixels have a fixed value    | Dark or bright border artifacts    |
| Replicate       | Repeat the closest boundary pixel    | Flat extension at the edge         |
| Reflect         | Mirror the image across the boundary | Often smoothest for natural images |
| Wrap            | Opposite edges are adjacent          | Appropriate for periodic data      |
| Valid only      | Skip incomplete neighborhoods        | Smaller output image               |

*i* Note

Two implementations using the same kernel can disagree near the boundary if their padding modes differ.

# Common Output Modes

For an  $M \times N$  image and a  $P \times Q$  kernel:

| Mode               | Output size                      | Interpretation                         |
|--------------------|----------------------------------|----------------------------------------|
| <code>full</code>  | $(M + P - 1) \times (N + Q - 1)$ | Every partial overlap, with padding    |
| <code>same</code>  | $M \times N$                     | Center portion matching the input size |
| <code>valid</code> | $(M - P + 1) \times (N - Q + 1)$ | Only complete overlaps; no padding     |

Example: convolving a  $275 \times 175$  image with another  $275 \times 175$  image produces:

- `full`:  $549 \times 349$
- `same`:  $275 \times 175$
- `valid`:  $1 \times 1$

 Note

Library defaults differ. Specify the mode and boundary behavior explicitly when reproducibility matters.

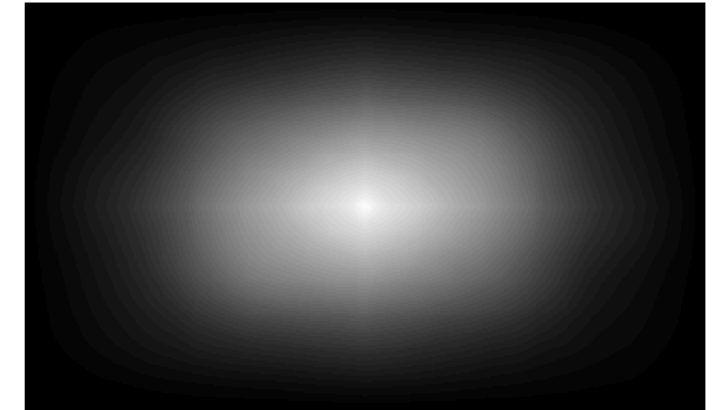
# Three Common Output Modes

```
1 import cv2
2
3 x = cv2.imread("assets/03_filtering/colorado")
4 x = cv2.resize(x, (200,120), interpolation=cv2.INTER_LINEAR)
5 h, w = x.shape
6
7 p = cv2.copyMakeBorder(x, h-1,0,w-1,0, cv2.BORDER_REPLICATE)
8 full = cv2.filter2D(p, cv2.CV_32F, x, anchor=cv2.Anchor(0,0,0,0))
9 same = cv2.filter2D(x, cv2.CV_32F, x, borderMode=cv2.BORDER_REPLICATE)
10 valid = full[h-1:h, w-1:w]
11
12 ims, names = [x,full,same,valid], ["Input","Full","Same","Valid"]
13 H, W = full.shape
14 fig, ax = plt.subplots(2,2,figsize=(6.5,5))
15
16 for a, im, name in zip(ax.flat, ims, names):
17     ih, iw = im.shape
18     a.imshow(im, cmap="gray", extent=(0,iw,ih,0))
19     a.set(xlim=(0,W), ylim=(H,0), title=fr"{name}")
20     a.axis("off")
21
22 plt.tight_layout(pad=.4)
```

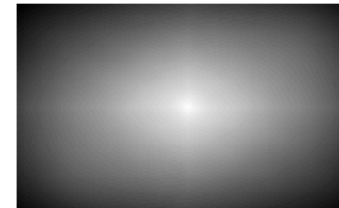
Input: 200 × 120



full: 399 × 239



same: 200 × 120



valid: 1 × 1

# Practice with Linear Filters

Predict the effect of each kernel before viewing the following examples.

Input



Kernel

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 0 | 0 | 0 |

Kernel

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 0 | 1 |
| 0 | 0 | 0 |

Kernel

|   |   |    |
|---|---|----|
| 1 | 0 | -1 |
| 2 | 0 | -2 |
| 1 | 0 | -1 |

Kernel

|      |      |      |
|------|------|------|
| -1/9 | -1/9 | -1/9 |
| -1/9 | 8/9  | -1/9 |
| -1/9 | -1/9 | -1/9 |

# Practice:

What will this kernel do to the image?

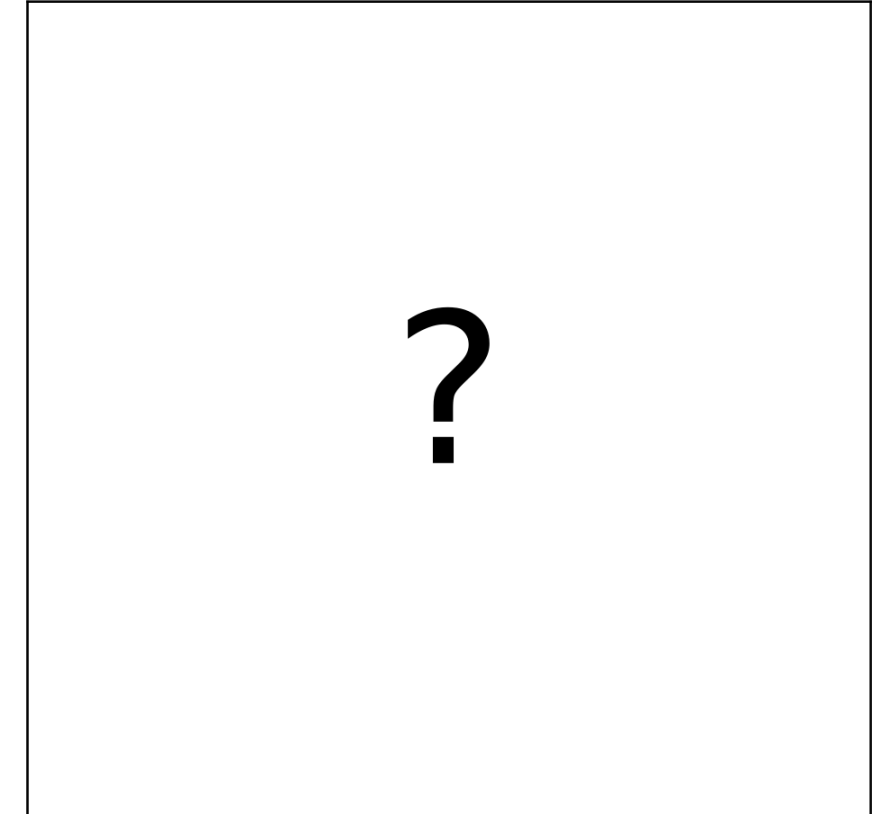
Input



Kernel

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 0 | 0 | 0 |

Predict the output



# Identity Kernel Leaves the Image Unchanged

The center coefficient copies the current pixel:

$$y[m, n] = I[m, n].$$

Input



Kernel

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 0 | 0 | 0 |

Output: unchanged





# Practice:

What will happen when the kernel selects the pixel immediately to the right?

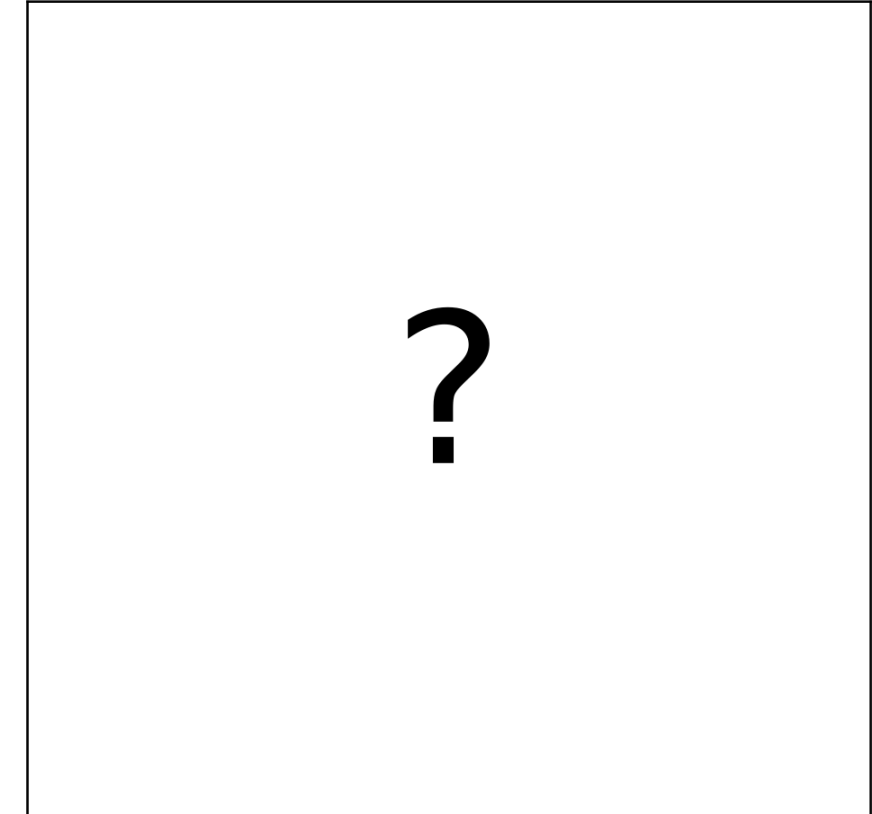
Input



Kernel

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 0 | 1 |
| 0 | 0 | 0 |

Predict the output





# Shift Kernel Copies a Neighboring Pixel

For correlation, the kernel produces

$$y[m, n] = I[m, n + 1],$$

so the image content shifts left by one pixel.

Input



Kernel

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 0 | 1 |
| 0 | 0 | 0 |

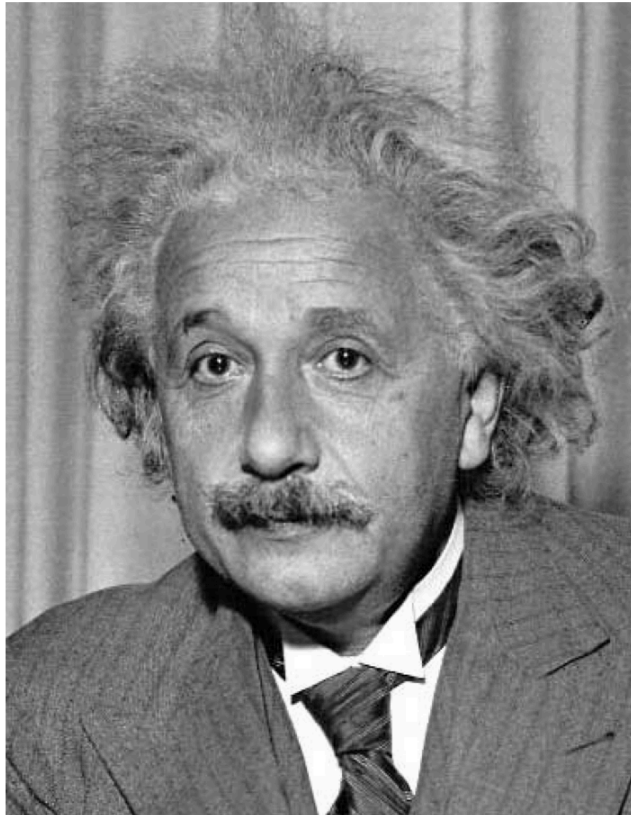
Output: shifted left



# Practice:

Examine the kernel and predict which image structures will produce the strongest response.

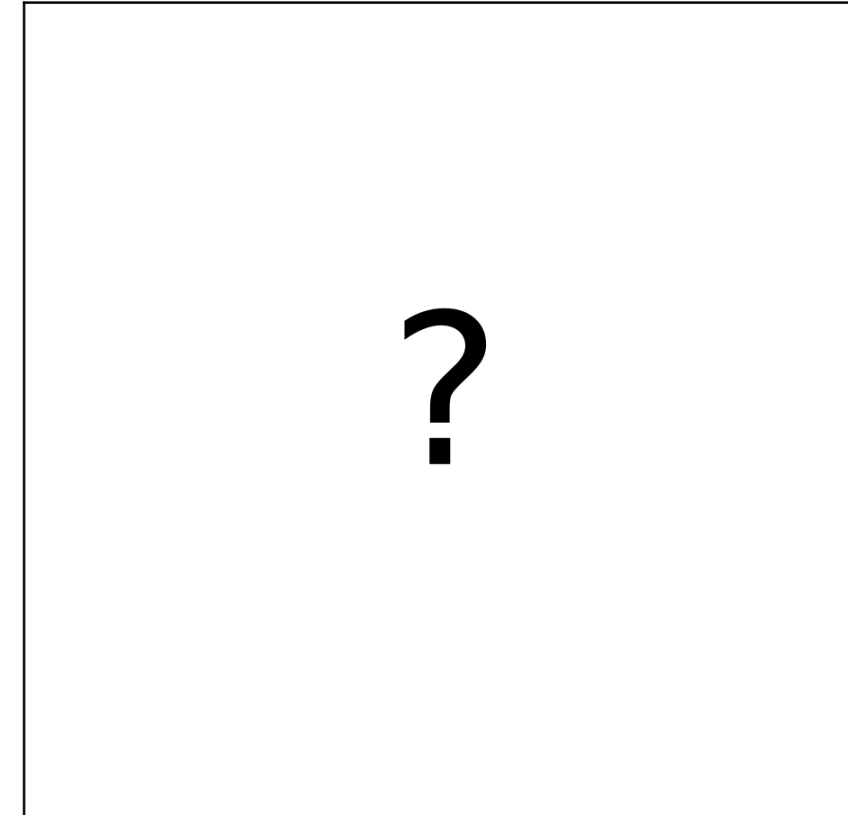
Input



Kernel

|   |   |    |
|---|---|----|
| 1 | 0 | -1 |
| 2 | 0 | -2 |
| 1 | 0 | -1 |

Predict the output

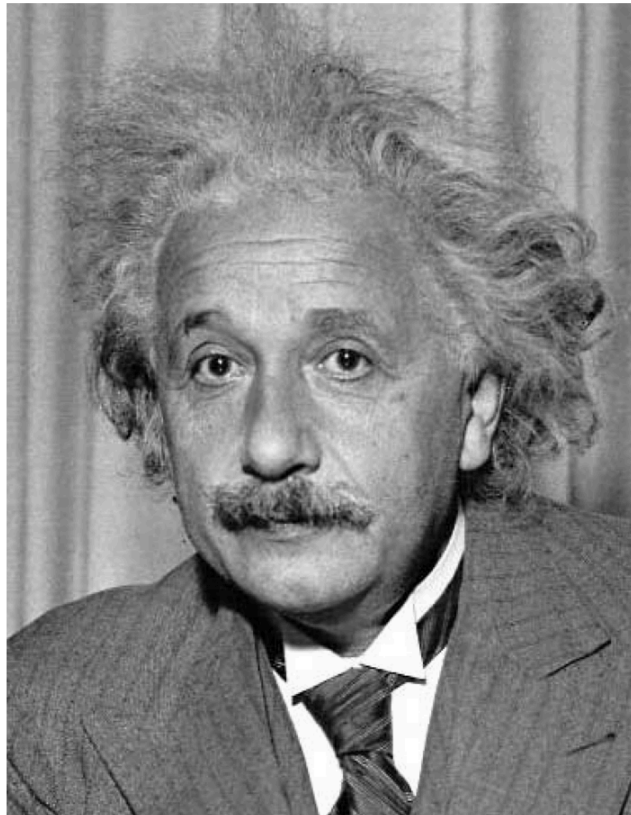


- Will the kernel respond to horizontal or vertical edges?
- Which regions of the image will have a response near zero?

# Sobel $x$ : Detecting Vertical Edges

This kernel responds strongly to vertical edges.

Input



Kernel

|   |   |    |
|---|---|----|
| 1 | 0 | -1 |
| 2 | 0 | -2 |
| 1 | 0 | -1 |

Absolute response

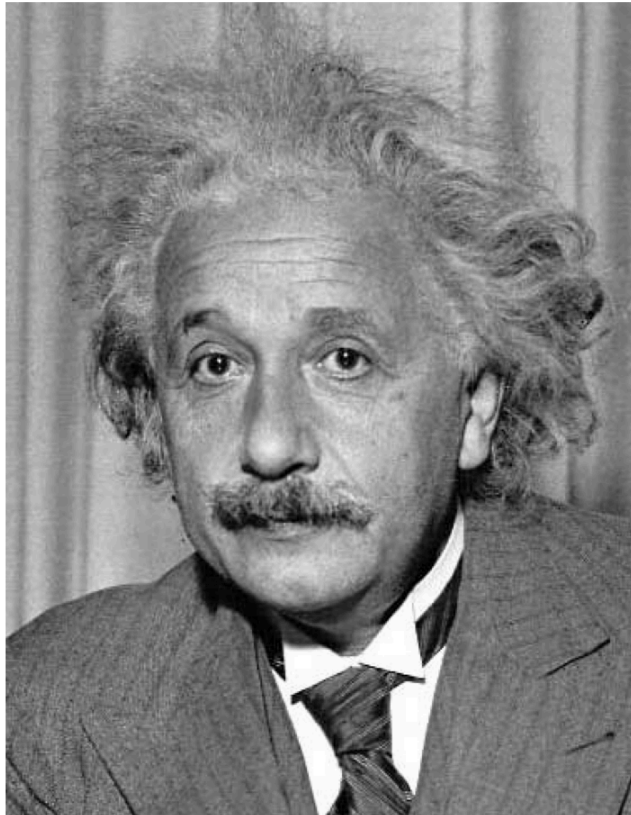


The absolute value displays both dark-to-light and light-to-dark transitions as bright edges.

# Practice:

Examine the kernel and predict which image structures will produce the strongest response.

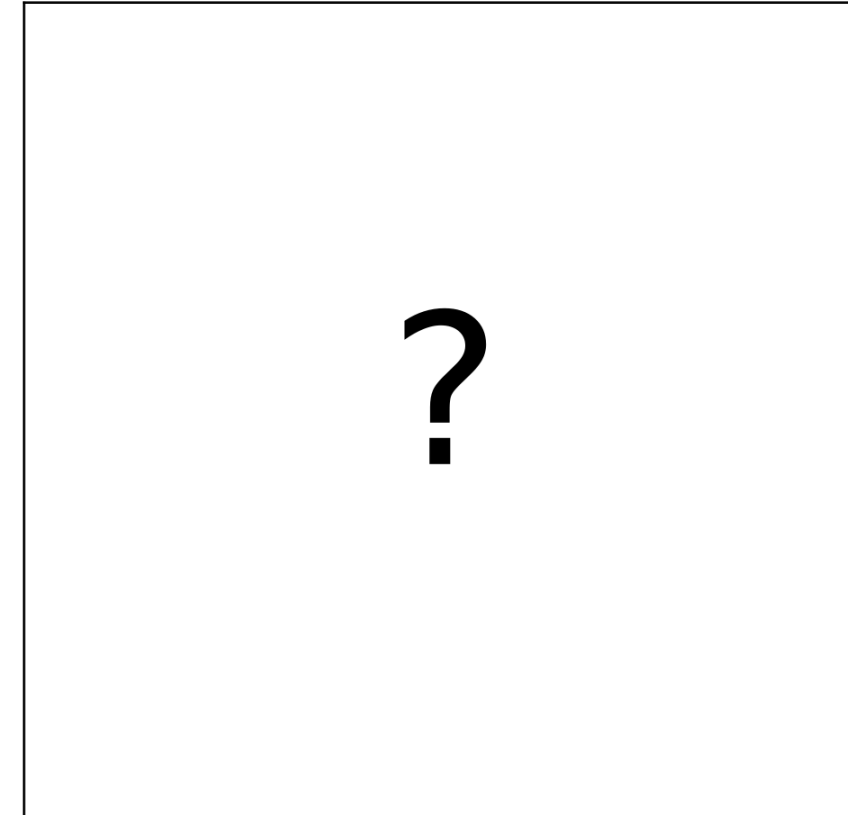
Input



Kernel

|    |    |    |
|----|----|----|
| 1  | 2  | 1  |
| 0  | 0  | 0  |
| -1 | -2 | -1 |

Predict the output

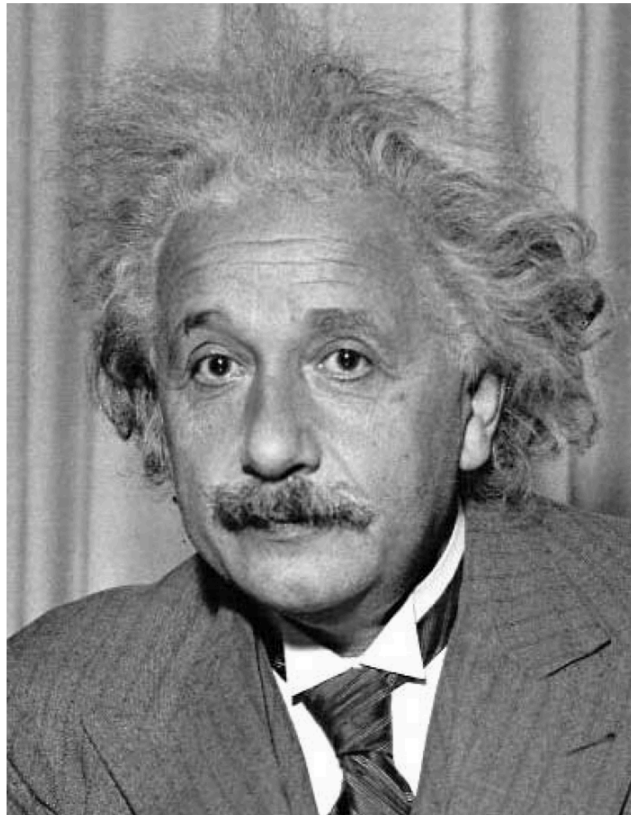


- Will the kernel respond to horizontal or vertical edges?
- Which regions of the image will have a response near zero?

# Sobel $y$ : Detecting Horizontal Edges

This kernel responds strongly to horizontal edges.

Input



Kernel

|    |    |    |
|----|----|----|
| 1  | 2  | 1  |
| 0  | 0  | 0  |
| -1 | -2 | -1 |

Absolute response



The sign of the original response indicates the direction of the intensity transition.

# Practice: Subtracting the Local Average

What remains after subtracting the local  $3 \times 3$  average from the center pixel?

$$h_{\text{detail}} = \delta - h_{\text{box}}.$$

Input



Kernel

|      |      |      |
|------|------|------|
| -1/9 | -1/9 | -1/9 |
| -1/9 | 8/9  | -1/9 |
| -1/9 | -1/9 | -1/9 |

Predict the output

?



# Local-Detail Kernel Extracts High Frequencies

The kernel removes smooth content and preserves rapid intensity changes:

$$y = I - \text{boxblur}(I).$$

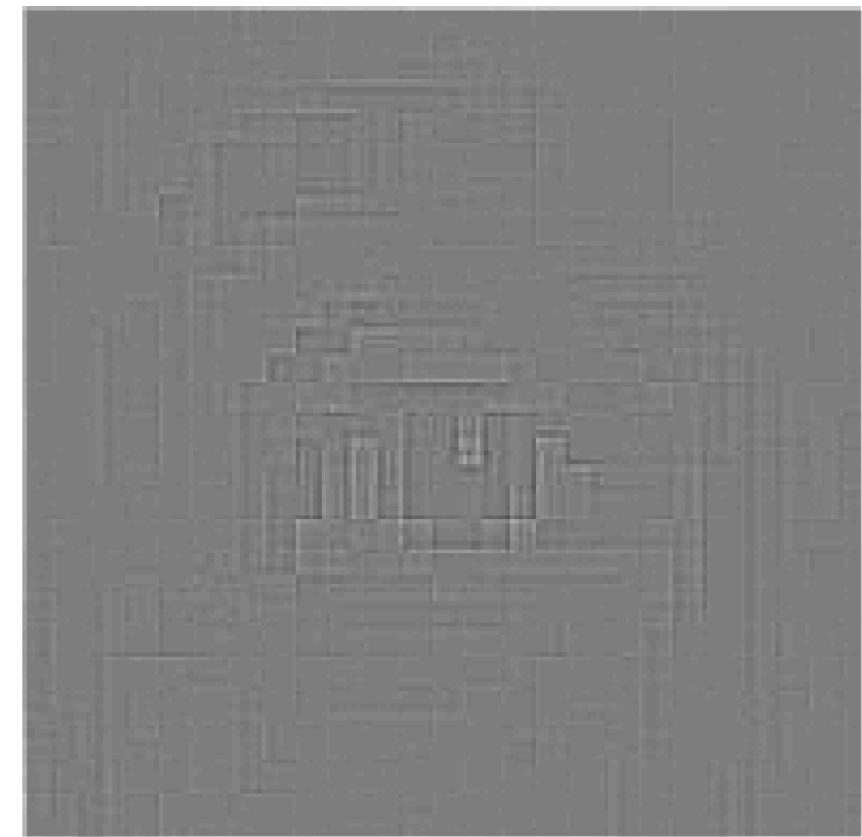
Input



Kernel

|      |      |      |
|------|------|------|
| -1/9 | -1/9 | -1/9 |
| -1/9 | 8/9  | -1/9 |
| -1/9 | -1/9 | -1/9 |

Signed local detail



Mid-gray represents weak response; brighter or darker values represent strong filter response.

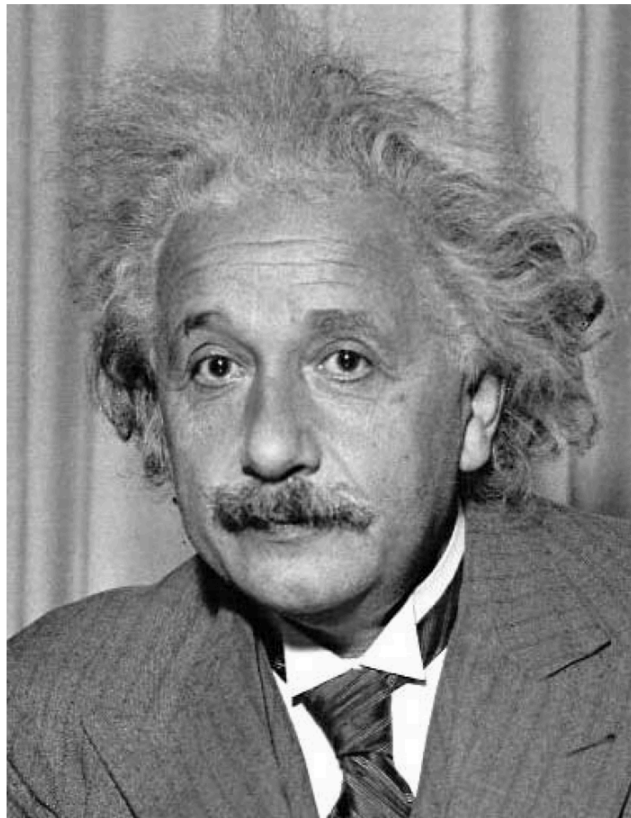
# Adding Local Detail Sharpens the Image

Sharpening adds the extracted high-frequency detail back to the original image:

$$h_{\text{sharp}} = \delta + (\delta - h_{\text{box}})$$

$$y = I + (I - \text{boxblur}(I))$$

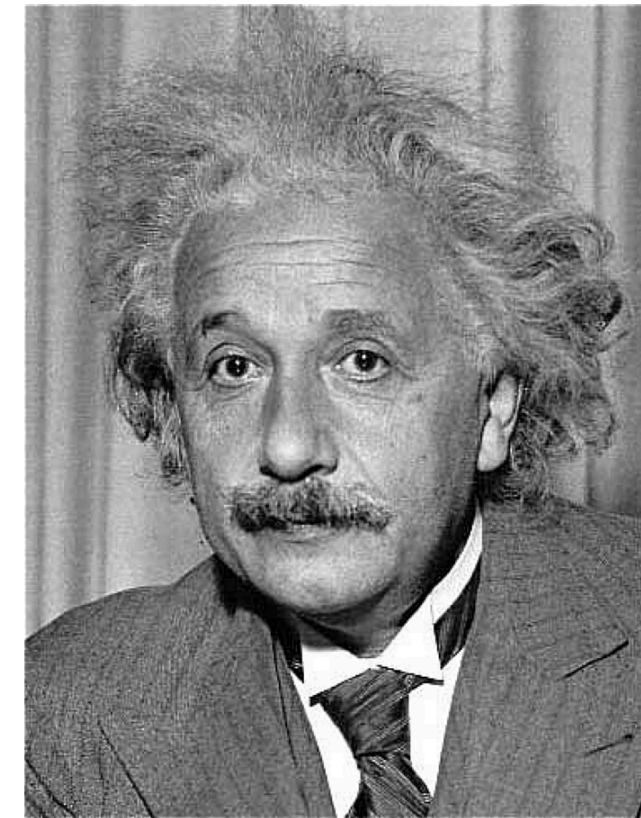
Input



Kernel

|      |      |      |
|------|------|------|
| -1/9 | -1/9 | -1/9 |
| -1/9 | 17/9 | -1/9 |
| -1/9 | -1/9 | -1/9 |

Sharpened output





# Filter Responses Can Be Negative

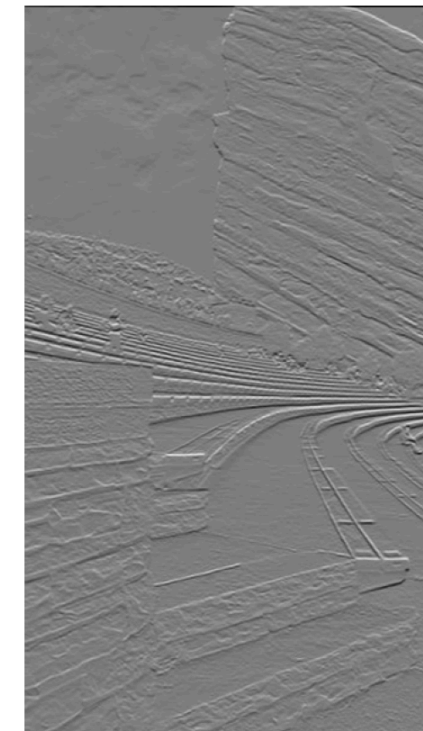
- Consider the horizontal-edge Sobel kernel  $h$ .
- A Sobel filter estimates an image derivative, so output is not restricted to the range  $[0, 255]$  or  $[0, 1]$ .
- Positive and negative values represent opposite edge polarities.
- Values near zero indicate little intensity change.

$$h = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

```
1 img = cv2.imread("assets/03_filtering/red-rocks.jpg")
2 h_y = np.array([[1, 2, 1],
3                 [0, 0, 0],
4                 [-1, -2, -1]], dtype=np.float32)
5 img_out = cv2.filter2D(img, cv2.CV_32F, h_y,
6 limit = np.abs(img_out).max())
7
8 fig, ax = plt.subplots(1, 2, figsize=(6.5, 4))
9 ax[0].imshow(img, cmap="gray"); ax[0].set_title("Input")
10 ax[1].imshow(img_out, cmap="gray", vmin=-limit, vmax=limit)
11 ax[1].set_title("Signed filter response")
12 for a in ax: a.axis("off")
13 plt.tight_layout(); plt.show()
```



Signed filter response



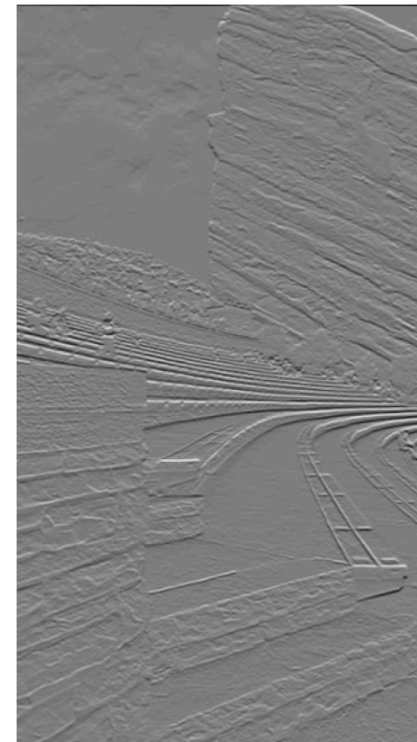
# Visualizing Signed Filter Responses

The signed filter response can be transformed differently depending on what we want to visualize.

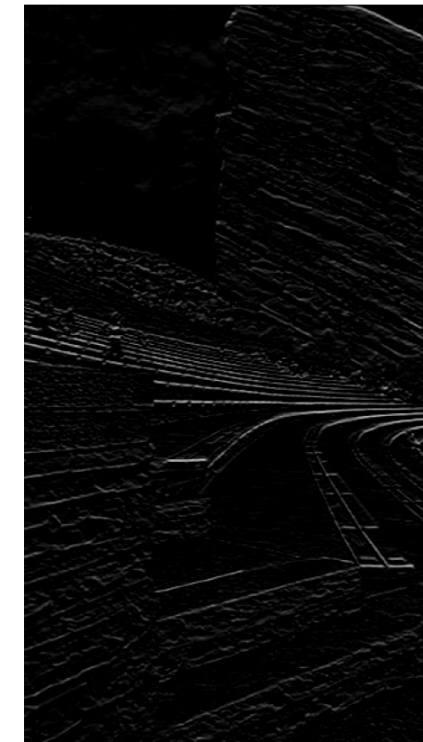
- In the signed response, zero is centered at gray.
- Clipping negative values removes one edge polarity.
- The absolute response shows edge strength but discards polarity.

```
1 limit = max(float(np.abs(img_out).max()), 1)
2
3 images = [img_out, np.clip(img_out, 0, None)]
4
5 titles = ["Signed", "Neg clipped", "Absolute"]
6 fig, ax = plt.subplots(1, 3, figsize=(6.5, 4))
7 for a, image, title in zip(ax, images, titles):
8     signed = title == "Signed"
9     a.imshow(image, cmap="gray", vmin=-limit if signed else 0)
10    a.set_title(title); a.axis("off")
11 plt.tight_layout(); plt.show()
```

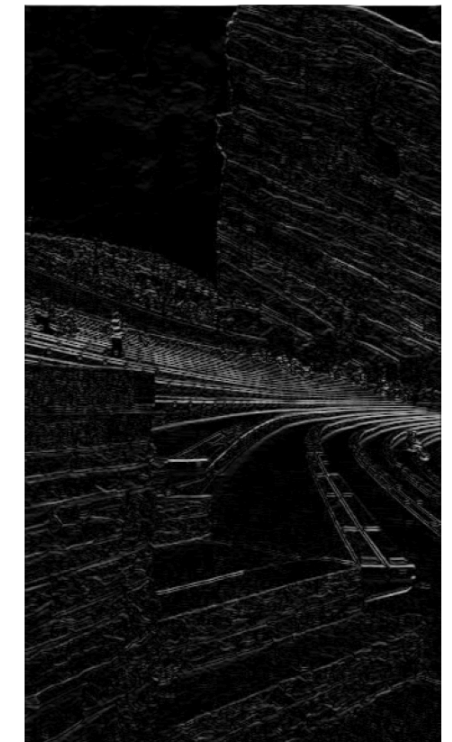
Signed



Neg clipped



Absolute val



# Practice: Match Each Filter to Its Output

Select E, F, G, H, or I for each filter. One candidate is not used.

1.  $A * B = ?$       2.  $B * C = ?$       3.  $A * D = ?$       4.  $A * A = ?$

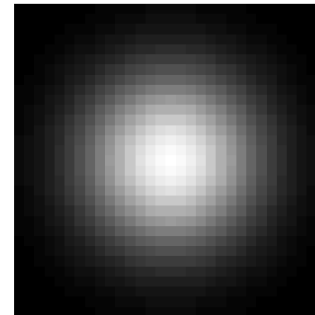
A: input image



B: derivative kernel



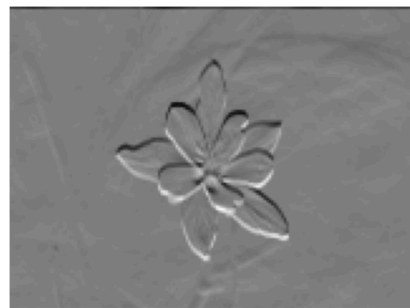
C: Gaussian kernel



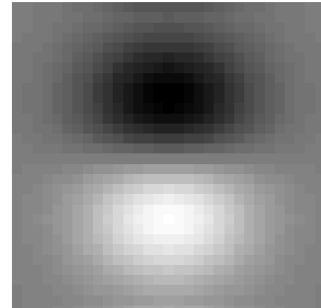
D: shifted impulse



E



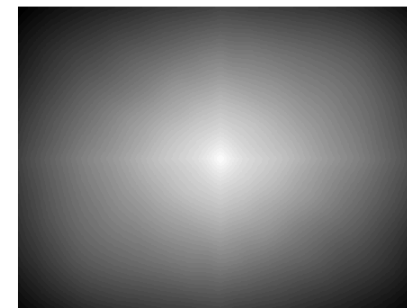
F



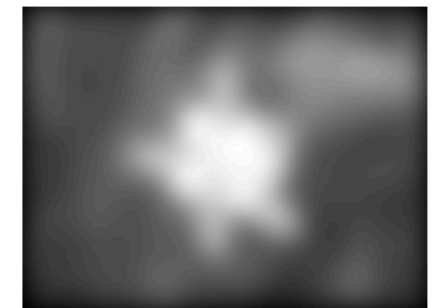
G



H

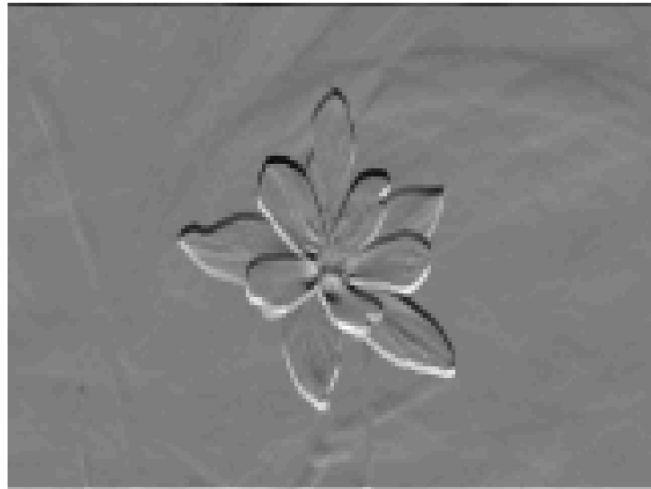


I

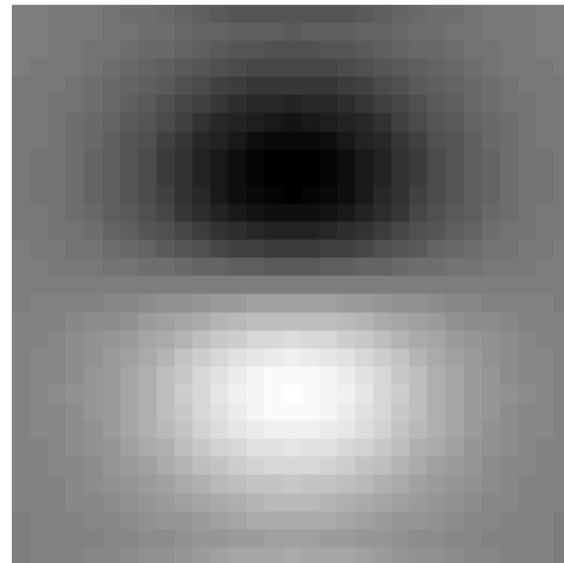


# Practice: Answers

$$A * B = E$$



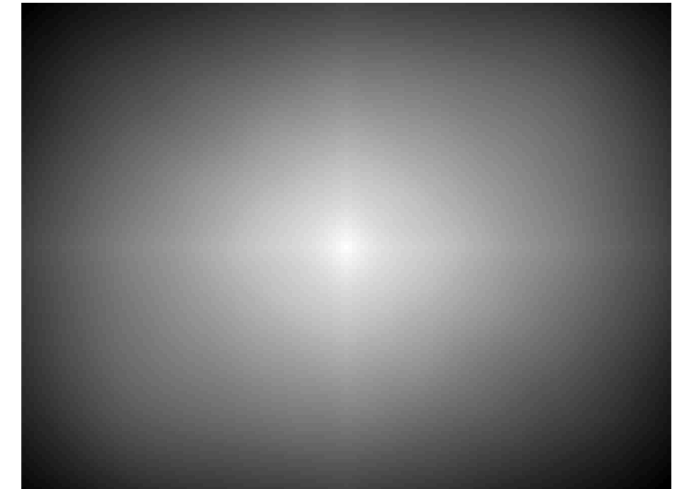
$$B * C = F$$



$$A * D = G$$



$$A * A = H$$



- $A * B = E$ : the derivative kernel produces an edge response.
- $B * C = F$ : smoothing the derivative kernel produces a derivative-of-Gaussian kernel.
- $A * D = G$ : correlation with the shifted impulse shifts the image.
- $A * A = H$ : correlation the image with itself produces a broad self-correlation response.

The unused candidate is  $I = A * C$ , the Gaussian-smoothed image.

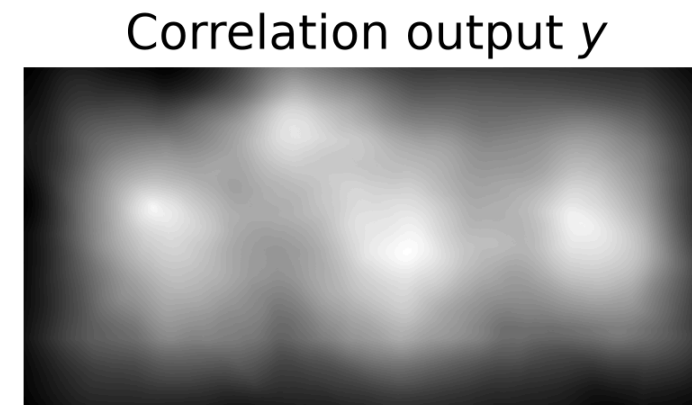
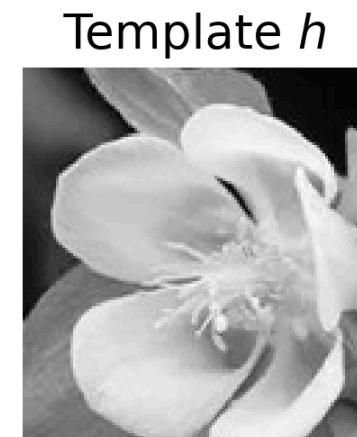
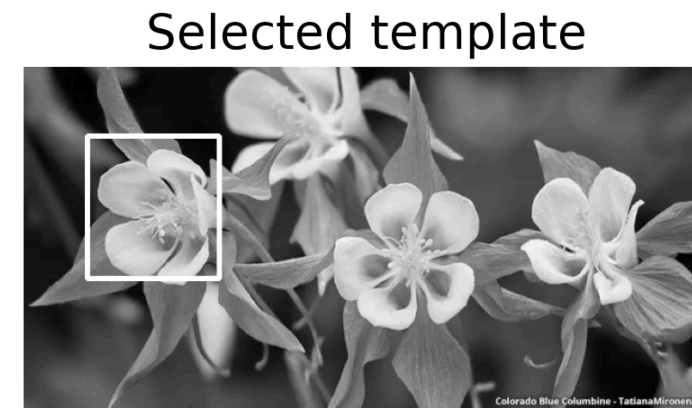
# Correlation as Template Matching

- Use an image patch as the filter kernel and correlate it across the image:

$$y[m, n] = \sum_k \sum_l h[k, l] I[m + k, n + l]$$

- The largest response is the location that best matches the template.

```
1 import cv2
2
3 path = "assets/03_filtering/colorado-flowers
4 box = (60, 65, 125, 135) # x0, y0, width, height
5
6 I = cv2.imread(path, cv2.IMREAD_GRAYSCALE).astype(np.float32)
7 x0, y0, w, ht = box
8 h = I[y0:y0+ht, x0:x0+w]
9
10 y = cv2.filter2D(I, cv2.CV_32F, h, borderType=cv2.BORDER_REPLICATE)
11 _, score, _, (px, py) = cv2.minMaxLoc(y)
12
13 selected = I.copy(); cv2.rectangle(selected, (x0, y0), (x0+w, y0+ht),
14 match = I.copy()
```





# Brightness Can Dominate Correlation

- Raw correlation computes a dot product, so brighter regions can produce larger responses.
- Subtract the template mean:  $f_0 = f - \bar{f}$
- Now positive and negative values describe the **pattern relative to its local brightness**.

```
1 h0 = h - h.mean()
2 y = cv2.filter2D(I, cv2.CV_32F, h0, borderType=cv2.BORDER_REPLICATE)
3
4 _, score, _, (px,py) = cv2.minMaxLoc(y)
5 match = I.copy()
6 cv2.rectangle(match, (px-w//2,py-h//2), (px+w//2,py+h//2), color=(0,0,255), thickness=2)
7
8 fig, ax = plt.subplots(2,2,figsize=(6.5,4.5))
9 ax[0,0].imshow(h0,cmap="gray"); ax[0,0].set_title("Zero-mean h")
10 ax[1,0].imshow(y,cmap="gray"); ax[1,0].plot(px,py, "r.")
11 ax[1,0].set_title("Correlation output y")
12 ax[1,1].imshow(match,cmap="gray"); ax[1,1].set_title("Peak: (122, 132)")
13 for a in ax.flat: a.axis("off")
14 plt.tight_layout(pad=.3); plt.show()
```

Zero-mean  $h$



Correlation output  $y$



Peak: (122, 132)



# Use Correlation, Not Convolution, for Template Matching

- **Correlation** compares the template in its original orientation.
- **Convolution** compares against a template rotated by  $180^\circ$ .
- For **template matching**, use **correlation** so the template orientation is preserved.

```
1 kernels = [h0, cv2.flip(h0, -1)]
2 names = ["Correlation", "Convolution"]
3
4 fig, ax = plt.subplots(2, 2, figsize=(6.5, 4.4))
5
6 for r, (hk, name) in enumerate(zip(kernels, names)):
7     y = cv2.filter2D(I, cv2.CV_32F, hk, borderType=cv2.BORDER_REPLICATE)
8     _, score, _, (px, py) = cv2.minMaxLoc(y)
9
10    match = I.copy()
11    cv2.rectangle(match, (px-w//2, py-h//2), (px+w//2, py+h//2), (0, 255, 0, 2))
12
13    ax[r, 0].imshow(hk, cmap="gray")
14    ax[r, 1].imshow(match, cmap="gray")
15    ax[r, 0].set_title(f"{name} kernel")
16    ax[r, 1].set_title(f"Peak response = {score}")
17
```

Correlation kernel



Convolution kernel



Peak response = 982.2



Peak response = 583.8



# Nonlinear Filters

$$y = \text{filter}(I, h)$$

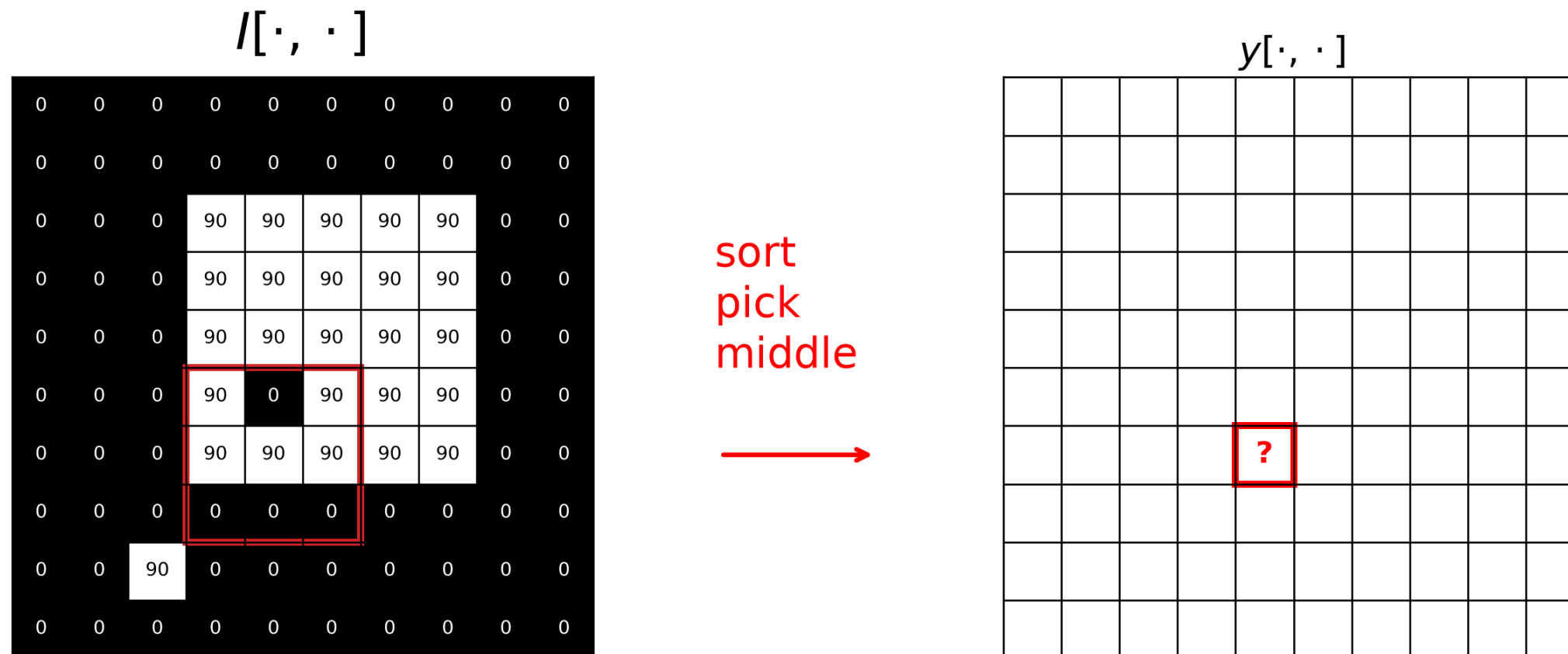


# Median Filter

- A median filter is a nonlinear **rank filter**:

$$y[m, n] = \text{median}\{I[m + k, n + l]\}_{(k,l) \in \Omega}.$$

- It replaces the center pixel by the middle value in the local neighborhood.
- Unlike a linear filter, there is **no kernel**  $h[k, l]$  such that  $y[m, n] = \sum_k \sum_l h[k, l] I[m + k, n + l]$ .



# Salt-and-Pepper Noise: Mean vs. Median

- Impulse noise replaces some pixels with 0 or 255.
- A mean filter averages these outliers with their neighbors, while a median filter can reject them.

```
1 path = "assets/03_filtering/colorado-flower.  
2 I0 = cv2.imread(path, cv2.IMREAD_GRAYSCALE)  
3 I0 = cv2.resize(I0, (380,255))  
4  
5 # Add salt-and-pepper noise  
6 rng, p = np.random.default_rng(7), 0.03  
7 Isp = I0.copy(); M = rng.random(I0.shape)  
8 Isp[M < p] = 0; Isp[M > 1-p] = 255  
9  
10 # Compare mean and median filtering  
11 Imean3 = cv2.blur(Isp, (3,3), borderType=cv2  
12 Imed3 = cv2.medianBlur(Isp, 3)  
13  
14 fig, ax = plt.subplots(2,2,figsize=(6.5,4.4)  
15 for a, im, t in zip(ax.flat,  
16     [I0, Isp, Imean3, Imed3],  
17     ["Original $I$", "Salt-and-pepper noise"  
18     panel(a, im, t)  
19 plt.tight_layout(pad=.3); plt.show()
```

Original /



Salt-and-pepper noise



3x3 mean



3x3 median



# Window Size Matters

- Larger windows remove more noise, but they also remove more image detail.

```
1 # Mean filters
2 Imean3 = cv2.blur(Isp, (3,3), borderType=
3 Imean11 = cv2.blur(Isp, (11,11), borderType=
4
5 # Median filters
6 Imed3 = cv2.medianBlur(Isp, 3)
7 Imed11 = cv2.medianBlur(Isp, 11)
8
9 fig, ax = plt.subplots(2,2,figsize=(6.5,4.4)
10 for a, im, t in zip(ax.flat,
11     [Imean3, Imean11, Imed3, Imed11],
12     ["3×3 mean", "11×11 mean", "3×3 median",
13     panel(a, im, t)
14 plt.tight_layout(pad=.3); plt.show()
```

3×3 mean



11×11 mean



3×3 median



11×11 median



# Median Filters

- The median filter operates over a window by selecting the median intensity in the window.
- Because ordering is nonlinear, in general,

$$\text{median}(I_1 + I_2) \neq \text{median}(I_1) + \text{median}(I_2)$$

## Note

- What advantage does a median filter have over a mean filter?
- Is a median filter a kind of convolution?

# Filtering Applications

## Improve an image:

- Remove noise
- Smooth before downsampling
- Sharpen local detail
- Estimate illumination or background

## Extract information:

- Detect edges and corners
- Measure texture
- Find repeated patterns
- Match a template to image regions

| Goal                 | Useful starting point       | Main tradeoff                   |
|----------------------|-----------------------------|---------------------------------|
| Reduce noise         | Gaussian smoothing          | Blurs fine details              |
| Remove impulse noise | Median filtering            | Can remove small structures     |
| Detect edges         | Sobel or derivative kernels | Amplifies noise                 |
| Enhance local detail | Sharpening / high-pass      | May create noise or halos       |
| Find a known pattern | Normalized correlation      | Sensitive to scale and rotation |

# Key Takeaways

- A filter maps each output location to a function of a local input neighborhood.
- A kernel's weights determine whether the filter smooths, differentiates, shifts, or sharpens.
- Correlation uses the kernel as written; convolution rotates it by  $180^\circ$ .
- Gaussian filters are composable and separable.
- Sobel filters estimate signed spatial derivatives and edge strength.
- Boundary rules and output modes are part of the algorithm.
- Median filtering is nonlinear and robust to isolated extreme values.